

CÆNL: Collapse-Based Active Neural Learning with Objective-Aligned Representation Control

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ARTICLE INFO

Keywords:

neural collapse
active learning
representation geometry
feedback control
regularization
discriminative learning

ABSTRACT

Representation regularization can alter feature geometry without improving prediction. We study objective-aligned representation control within an active-learning protocol. Collapse-Based Active Neural Learning (CÆNL) separates acquisition from Dynamic Collapse Regularization (DCR), which penalizes a normalized within-class to between-class scatter ratio and includes a separation safeguard. We compare fixed strength, bounded deterministic feedback (MACC-Lite), and learned regularization control (Full MACC). Supporting analysis characterizes the scatter statistic, an isolated regularizer step, coefficient bounds, and a conditional nearest-centroid error bound. Across 12 ImageNet-100 conditions and six paired seeds, entropy acquisition with aligned DCR and MACC-Lite reaches $73.360 \pm 0.208\%$ accuracy (mean $\pm$ seed SD) at 60% annotations. Its gain over entropy alone is 1.897 percentage points, with an individual 95% paired- t interval of [1.625, 2.169]. This follow-up comparison passes its within-family Holm-adjusted paired- t test. A later matched fixed-strength control reaches $72.263 \pm 0.381\%$; Lite exceeds it in every seed, by 1.097 points on average with interval [0.714, 1.479]. Both contrasts in this new two-comparison family pass the adjusted paired- t test, while the six-seed adjusted exact tests cannot reach the rejection threshold. This isolates a setting-specific advantage over the specified fixed coefficient, without establishing superiority to a tuned fixed schedule. Full MACC shows no additional classification benefit, and robust accuracy under AutoAttack at 8/255 is near zero. Separate spectral-regularization studies find a small GPT-2/C4 loss reduction with fixed DCR, worse CIFAR-10 diffusion FID, and no established AudioCaps CIDEr-D improvement. The findings support a specific classification treatment and a favorable Lite-versus-fixed contrast, while delineating the limits of mechanism attribution and cross-task scope.

1. Introduction

Neural collapse describes geometric regularities observed during the terminal phase of classifier training: within-class variability contracts, centered class means approach simplex structure, classifier weights align with those means, and decisions approach a nearest-centroid rule [1]. These observations motivate interventions in representation space. They do not imply that every reduction in rank or dispersion improves prediction, acquisition, or input-space robustness.

Active learning adds nonstationarity because new annotations change class support, optimization dynamics, and the information available for estimating geometry. Conventional acquisition criteria emphasize predictive uncertainty, coverage, or gradient diversity [2–4]. Representation-based acquisition can complement such criteria, but a geometrically atypical example may also be redundant, mislabeled, or out of distribution. Its deviation is not, by itself, a calibrated probability of epistemic uncertainty.

We study Collapse-Based Active Neural Learning (CÆNL), a framework that separates label acquisition from representation regularization. Here, collapse-based refers to controlling within-class dispersion relative to between-class separation; it does not assume that arbitrary rank reduction improves prediction. Acquisition may use a predicted-class geometry score or an established criterion such as entropy. For classification, our design principle is *objective–feedback*

alignment: the scatter statistic observed by the regularization controller also appears explicitly in the modulated loss. We investigate this design rather than assume it is necessary or sufficient for useful feedback. The aligned objective uses normalized within-class and between-class scatter. A separate spectral objective is retained for language modeling, unconditional image generation, and audio captioning, where the supervised class-partition objective is not applied.

The central question is whether regulating this statistic contributes to classification performance under a fixed acquisition and optimization protocol. Three comparisons organize the study: adding aligned DCR at fixed strength, replacing that strength with feedback, and replacing the acquisition rule while retaining the same loss and controller. The last comparison matters because an improvement in the complete pipeline need not be an improvement in acquisition itself.

Our contribution has three parts. First, we specify a normalized scatter objective with a separation safeguard and compare fixed, deterministic, and learned coefficient control. The construction combines established discriminative moments with explicit objective–feedback consistency; it does not introduce a new definition of neural collapse. Second, we give supporting mathematical characterizations and state the assumptions under which they apply. Third, we provide a paired-seed empirical study that distinguishes component comparisons from complete pipeline comparisons and retains unfavorable outcomes.

The strongest observed ImageNet-100 treatment combines entropy acquisition with aligned DCR and MACC-Lite. It improves mean accuracy by 1.897 percentage points

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over entropy alone at 60% annotations. The original matrix provides a Lite-versus-fixed comparison under geometric acquisition. A subsequent entropy-plus-fixed control supplies the corresponding comparison within entropy acquisition: Lite improves over the fixed coefficient by 1.097 points. This supports the specified adaptive treatment relative to $\lambda_l = 0.03$, without isolating the benefit of feedback from every alternative coefficient schedule. The entropy-plus-Lite, UHerding, and entropy-plus-fixed additions are follow-up studies using the original six seeds, with separately declared statistical families (Section 6.1).

The language, diffusion, and audio studies examine a separate spectral formulation. Their mixed outcomes characterize its scope; they are not replications of the aligned classification objective. We retain the small C4 loss reduction, adverse diffusion results, inconclusive audio results, and near-zero classification robustness. The resulting evidence concerns task-dependent representation regularization under specified protocols.

Sections 2–5 give the motivation, method, and analysis. Sections 6 and 7 report the design and principal results. Section 8 discusses interpretation and limitations. The main text includes the task protocols, paired comparisons, and computational measurements. The appendices provide seed-level values, secondary C4 tests, and data-verification details.

2. Background and related work

2.1. Neural collapse and discriminative geometry

The original neural-collapse characterization distinguishes variability collapse, simplex structure, self-duality, and nearest-centroid behavior [1]. A trace ratio summarizes only part of that structure. It is not a complete neural-collapse test and is not interchangeable with a pseudoinverse-normalized NC1 statistic. Within/between-class scatter also has a classical discriminant-analysis interpretation [5]. The use of these moments is therefore not itself claimed as new. Center loss penalizes distances to learned class centers when combined with softmax training [6]. DeepLDA optimizes a discriminative objective on learned representations [7]. Our construction instead applies a one-sided penalty to the normalized log-scatter ratio, freezes a reference target, and compares coefficient controls that observe that same ratio. These differences specify the method; the present experiments do not establish superiority over center loss or DeepLDA.

Wang et al. [8] examine alignment between feature and classifier spaces in long-tailed learning. Their notion of alignment differs from the objective–feedback consistency studied here. Alcalá et al. [9] characterize non-simplex geometry in the orthoplex regime. These studies caution against a universal equiangular or low-rank interpretation. Our theoretical claims concern the specified moment objective, not the global optimality of neural collapse in arbitrary networks.

2.2. Acquisition and learned control

Entropy sampling ranks predictive uncertainty. Core-set methods select representative embeddings [3], while BADGE combines uncertainty and diversity through hallucinated last-layer gradient embeddings [4]. Noise Stability measures output variation under parameter perturbations [10]. Our Mahalanobis criterion instead measures deviation from the predicted class using statistics estimated from acquired examples. It is a roundwise geometry score, not a measured sample-specific temporal change in collapse.

Recent work considers adaptive acquisition and its computational cost. Jung et al. [11] study acquisition-function aggregation and accuracy–cost trade-offs. Cui et al. [12] use labeled-data feedback and reinforcement learning for batch acquisition. These are related approaches, not executed comparators in the present matrix. Their acquisition actions differ from the Full-MACC condition evaluated here, which controls regularization with its query-policy head disabled.

Uncertainty Herding [13] combines uncertainty and coverage in a fixed feature space. We evaluate its margin-weighted coverage mechanism through an independent implementation using shared-initial task-model features and the present candidate-pass and warm-start conventions. This is a numerical comparison under a common task-training protocol, not a reproduction of its published benchmark recipe.

2.3. Robustness and calibration

Adversarial evaluation requires an explicit input norm, perturbation radius, preprocessing pipeline, and attack procedure. We use standard AutoAttack [14] at L_∞ radius 8/255. Calibration is a separate outcome: expected calibration error (ECE) compares confidence and correctness within bins [15]. Neither calibration nor input-space robustness follows automatically from a favorable representation statistic.

3. Problem formulation and representation measurements

3.1. Sets, budgets, and evaluation

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ be a fixed training pool, $\mathcal{A}_r$ the examples annotated by round r , and $\mathcal{U}_r$ the remaining unlabeled pool. A feedback set $\mathcal{H} \subset \mathcal{A}_0$ is annotated within the budget but withheld from gradient updates. The gradient-training set is $\mathcal{T}_r = \mathcal{A}_r \setminus \mathcal{H}$. Official evaluation examples are separate. The annotation constraint is $|\mathcal{A}_r| \leq B$, including initial and feedback labels. Selection removes queried examples from $\mathcal{U}_r$ and does not access their hidden labels before selection.

The model f_θ produces logits and pooled intermediate features $z_l(x) \in \mathbb{R}^{d_l}$ at monitored layers $l \in \mathcal{L}$. Training, feedback, acquisition, and final evaluation use explicitly distinguished transforms. Performance is measured at fixed annotation budgets. A 60% budget alone does not establish a 40% annotation saving at equal predictive performance.

Table 1

Positioning relative to representative acquisition and representation methods. Numerical entries are ImageNet-100 clean accuracy at 60% annotations under this paper's matched protocol (mean $\pm$ seed SD), not numbers pooled from different published benchmarks. A dash denotes a conceptual comparison that was not evaluated here.

Method	Principal strength or focus	Distinction or limitation for this study	Matched top-1 (%)
Center loss / DeepLDA [6, 7]	Direct discriminative representation objectives	No matched coefficient-control experiment here	—
BADGE [4]	Uncertainty and gradient diversity	Acquisition criterion; does not modulate the scatter objective	71.110 ± 0.467
Noise Stability [10]	Perturbation-based acquisition signal	Higher measured acquisition cost in this implementation	71.023 ± 0.249
Uncertainty Herding [13]	Uncertainty-weighted feature coverage	Local fixed-initial-R50 adaptation; published benchmark recipe not reproduced	71.747 ± 0.626
Aggregation Set [11, 12]	Trust- Adaptive acquisition using feedback	Acquisition actions differ from the regularization-only Full MACC studied here	—
Aligned DCR + Lite (entropy)	Shared statistic in loss and coefficient feedback	Benefit established against specified controls; tuned-fixed superiority untested	73.360 ± 0.208

3.2. Class-balanced normalized scatter

For labeled features in a measurement batch, define

$$h_{i,l} = \frac{z_l(x_i)}{\max\{\|z_l(x_i)\|_2, \epsilon_n\}}. \quad (1)$$

For represented class c , let n_c be its count, $\mu_{c,l} = n_c^{-1} \sum_{i:y_i=c} h_{i,l}$, and $\bar{\mu}_l = K^{-1} \sum_{c=1}^K \mu_{c,l}$. The equal-class-weighted moments are

$$W_l = \frac{1}{K} \sum_{c=1}^K \frac{1}{n_c} \sum_{i:y_i=c} \|h_{i,l} - \mu_{c,l}\|_2^2, \quad (2)$$

$$B_l = \frac{1}{K} \sum_{c=1}^K \|\mu_{c,l} - \bar{\mu}_l\|_2^2, \quad (3)$$

$$q_l = \log(W_l + \epsilon_s) - \log(B_l + \epsilon_s), \quad \kappa_l = \exp(q_l). \quad (4)$$

Training batches contain $K = 16$ classes and eight examples per class. Assessment uses all represented classes with equal class weights, including when their counts differ. A smaller q_l indicates lower normalized within-class dispersion relative to between-class dispersion. It does not independently determine every pairwise class distance, the complete covariance spectrum, or the classifier boundary.

Unit normalization removes common feature-scale dependence away from the norm floor and preserves invariance to an orthogonal coordinate change. It also changes the geometry relative to unnormalized activations. Training maintains moving averages of q_l and B_l ; their temporal changes are aggregate control-state measurements, not sample-specific acquisition scores.

3.3. Sample-level geometry for acquisition

Acquisition uses *raw* pooled features rather than the normalized features of Equation (1). At each round, let $\hat{\mu}_{c,l}$

and $\hat{v}_{c,l,j}$ be the mean and diagonal population-variance estimates from acquired gradient-training examples of class c . With $\hat{c}(x) = \arg \max_c f_\theta(x)_c$, define

$$u_l(x) = \sum_{j=1}^{d_l} \frac{(z_{l,j}(x) - \hat{\mu}_{\hat{c}(x),l,j})^2}{\hat{v}_{\hat{c}(x),l,j} + \epsilon_a}, \quad (5)$$

$$s(x) = \sum_{l \in \mathcal{L}} \alpha_l u_l(x), \quad \alpha_l \geq 0, \quad \sum_l \alpha_l = 1. \quad (6)$$

Here $\epsilon_a = 10^{-5}$ and the two layer weights are 0.5. Statistics are reestimated at each acquisition checkpoint. Hidden pool labels do not enter the score. A high value measures predicted-class geometric deviation; a confident outlier may also receive a high score.

4. Objective-aligned DCR and regularization control

4.1. Frozen targets and classification objective

Each seed first produces a shared cross-entropy-only checkpoint. Targets are calibrated once from 16 fixed class-balanced anchor batches of acquired gradient-training images using deterministic center crops. For reference means $\bar{q}_{l,0}$ and $\bar{B}_{l,0}$, set

$$q_l^* = \bar{q}_{l,0} + \log(0.9), \quad B_l^{\min} = 0.5 \bar{B}_{l,0}. \quad (7)$$

The implemented calibration rejects a layer unless $\bar{B}_{l,0} > 10\epsilon_s$. Consequently, $B_l^{\min} > 5\epsilon_s > 0$, so the logarithm of the separation floor below is well defined. This is an initialization acceptance check, not a guarantee that between-class dispersion remains above the floor during training. The targets are shared across methods within a seed and remain fixed throughout acquisition. They are design choices

rather than theoretically optimal targets. The 16-class anchor target is not a directly comparable threshold for the 100-class assessment statistic.

Define the one-sided Huber penalty

$$H_\delta(u) = \begin{cases} 0, & u \leq 0, \\ u^2/(2\delta), & 0 < u < \delta, \\ u - \delta/2, & u \geq \delta. \end{cases} \quad (8)$$

The aligned classification objective is

$$R_l = H_\delta(q_l - q_l^*), \quad (9)$$

$$F_l = H_\delta(\log B_l^{\min} - \log(B_l + \epsilon_s)), \quad (10)$$

$$\mathcal{L}_{AD} = \mathcal{L}_{CE} + \sum_{l \in \mathcal{L}} \lambda_l^{\text{eff}} R_l + \beta_B \sum_{l \in \mathcal{L}} F_l. \quad (11)$$

We use $\delta = 0.25$, $\epsilon_s = \epsilon_n = 10^{-6}$, and $\beta_B = 0.01$. The floor penalty remains active when a policy disables a ratio term. It discourages vanishing aggregate between-class dispersion but neither excludes every degenerate stationary point nor ensures separation of every class pair. The unregularized reference disables both penalties.

The loss and classification feedback use the same q_l . This consistency does not eliminate the effects of cross-entropy, other layers, augmentation, momentum, the separation penalty, or changing class support. The dense class-incidence implementation computes the moments in $O(nKd_l)$ operations per layer, with fixed $K = 16$, without forming a $d_l \times d_l$ covariance matrix.

4.2. Aligned MACC-Lite

Let $\hat{q}_{l,t}$ be the training moving average at controller event t , with per-step observation weight 0.05, and let $e_{l,t} = \hat{q}_{l,t} - q_l^*$. The bounded leaky rule is

$$u_{l,t} = \lambda_{\max} \tanh\left(\frac{[e_{l,t}]_+}{\delta}\right), \quad (12)$$

$$\lambda_{l,t+1} = (1 - \rho)\lambda_{l,t} + \rho u_{l,t}. \quad (13)$$

The settings are $\lambda_{\max} = 0.10$, $\rho = 0.20$, and $\lambda_{l,0} = 0.03$. Fixed aligned DCR uses $\lambda_l = 0.03$. Unlike an accumulating integral-like rule, Equation (13) decreases the coefficient when the positive residual disappears. Its coefficient bounds do not prove closed-loop stability or target convergence.

4.3. Aligned Full MACC

Full MACC uses a policy network with two 128-unit hidden layers. Its state includes each layer's smoothed log ratio, target deviation, normalized temporal change, current coefficient, and activation flag, together with progress and feedback cross-entropy. With $\hat{\kappa}_{l,t} = \exp(\hat{q}_{l,t})$ and $q_{l,t}^{\text{prev}}$ the stored moving average from the preceding completed controller window, the temporal feature is

$$d_{l,t} = \frac{|\hat{q}_{l,t} - q_{l,t}^{\text{prev}}| \hat{\kappa}_{l,t}}{\max\{\hat{\kappa}_{l,t}, 10^{-12}\}}. \quad (14)$$

The numerator is the wrapper's instability signal; the denominator is applied when constructing the policy state. Away from the numerical floor, $d_{l,t} = |\hat{q}_{l,t} - q_{l,t}^{\text{prev}}|$: it is an absolute log-ratio change, not a signed change or a time derivative. The stored value is initialized to the calibration reference and refreshed after each window's reward and next-action processing. No additional standardization is applied.

Actions comprise layer-wise increments from $\{-0.01, -0.005, 0, 0.005, 0.01\}$ and binary flags. Coefficients are projected to $[0, 0.10]$ and multiplied by the flags to obtain λ_l^{eff} .

An action applies during the next window of at most 100 optimizer steps. Let ℓ_t^H be feedback cross-entropy after that window, $s_H = \max\{0.1, \ell_{\text{phase},0}^H\}$ the fixed scale at the start of the acquisition phase, and Ψ_t the sum of unweighted ratio penalties and fixed-weight floor penalties evaluated from smoothed training moments. The reward is

$$\begin{aligned} r_t = & \text{clip}\left(\frac{\ell_{t-1}^H - \ell_t^H}{s_H}, -1, 1\right) \\ & + 0.1 \text{clip}(\Psi_{t-1} - \Psi_t, -1, 1) \\ & - 0.01 \frac{1}{|\mathcal{L}|} \sum_{l \in \mathcal{L}} \frac{\lambda_{l,t}^{\text{eff}}}{0.10}. \end{aligned} \quad (15)$$

The unweighted ratio potential prevents an apparent geometric reward obtained solely by switching off its loss multiplier. Feedback uses $\mathcal{H}$, not the official validation set. The terminal action is credited before the phase ends. A new phase initializes its own feedback-loss baseline rather than assigning acquisition and learning-rate-reset effects to the preceding action.

Uniform exploration decreases from probability 0.10 to 0.02. Otherwise actions are sampled from the policy's categorical distributions. Uniform exploratory actions are excluded from on-policy REINFORCE updates [16]. The advantage uses the previous moving reward baseline, which is updated afterward. Policy learning rate is 3×10^{-4} , the entropy coefficient is 0.001, and the reward-baseline observation weight is 0.05. The first reward initializes the baseline and has zero centered advantage.

The acquisition-policy head is disabled. Full MACC therefore denotes learned regularization control within the active-learning loop, not learned query thresholds or variable annotation budgets. The acquisition quantile remains 0.90; scoring weights and queried label counts are fixed. Online policy adaptation here is not meta-training across tasks.

4.4. Round-based active learning

Figure 1 shows the complete data, measurement, and control flow; Algorithm 2 specifies its execution order.

Acquisition takes place between training phases; regularization control operates within them. At round r , required predictions and features for the remaining pool are cached. Candidate passes contain at most 10,000 examples. Up to 10% of a pass are selected, excluding previously selected examples; passes repeat until the round budget is reached. The

Algorithm 1: Window-timed Full MACC regularization update

Require: Pending state/action (S_t, A_t) , previous feedback loss and potential, baseline b_t .

- 1: Train the model for the next window under A_t .
- 2: Measure feedback loss on $\mathcal{H}$ and training potential; compute r_t using Equation (15).
- 3: **if** the reward baseline is uninitialized **then**
- 4: Set $b_t \leftarrow r_t$.
- 5: **end if**
- 6: $a_t \leftarrow r_t - b_t$.
- 7: $b_{t+1} \leftarrow 0.95b_t + 0.05r_t$.
- 8: **if** A_t was sampled from the policy, not uniform exploration **then**
- 9: Update the policy using $a_t \nabla_{\phi} \log \pi_{\phi}(A_t | S_t)$ and its entropy term.
- 10: **end if**
- 11: Clear the pending action.
- 12: **if** the current training phase continues **then**
- 13: Form S_{t+1} ; sample bounded action A_{t+1} and store it as pending.
- 14: **end if**

Mahalanobis conditions use Equation (6) with deterministic tie handling. Random, entropy, BADGE, and Noise Stability use the same budget and candidate-pass conventions but incur different computational costs.

4.5. Spectral formulation for language, diffusion, and audio

The other tasks use the separate spectral objective

$$\mathcal{L}_{\text{spec}} = \mathcal{L}_{\text{task}} + \sum_{l \in \mathcal{L}} \lambda_l \|p_l - p_l^*\|_2^2. \quad (16)$$

This is spectrum matching, not a Frobenius penalty between full covariance matrices and not the supervised scatter loss.

For a centered feature matrix $Z_l \in \mathbb{R}^{n_l \times d_l}$, the implementation computes descending, nonnegative eigenvalues of $Z_l^\top Z_l / n_l$, using the smaller Gram matrix $Z_l Z_l^\top / n_l$ when appropriate. Let $k_l = \min(k, d_l)$. The first k_l eigenvalues are retained; when fewer values are available, the vector is padded with zeros. With this convention,

$$p_{l,j} = \frac{\sigma_{l,j}}{\sum_{a=1}^{k_l} \sigma_{l,a} + \epsilon_{\text{spec}}}, \quad j = 1, \dots, k_l, \quad \epsilon_{\text{spec}} = 10^{-8}. \quad (17)$$

The stabilizer means that the implemented vector is not exactly unit mass when the retained spectral sum is very small. For an exactly normalized nonzero spectrum π , effective rank is $\exp(-\sum_j \pi_j \log \pi_j)$. The monitor uses the full moving covariance spectrum rather than the current truncated batch spectrum. Neither measurement is the supervised trace ratio of Equation (4).

A rank- r flat target places mass $1/r$ on the first r entries. The implementation restricts target rank to the available

Algorithm 2: The evaluated CÆNL active-learning protocol

Require: Pool $\mathcal{D}$, initial annotations $\mathcal{A}_0$, feedback set $\mathcal{H}$, round budget b , five rounds, monitored layers $\mathcal{L}$.

- 1: Train a shared cross-entropy initial model on $\mathcal{A}_0 \setminus \mathcal{H}$.
- 2: Calibrate and freeze targets using acquired training anchors.
- 3: **for** each compared method **do**
- 4: Restore the same initial model and optimizer; initialize its controller.
- 5: $\mathcal{A} \leftarrow \mathcal{A}_0$; $\mathcal{U} \leftarrow \mathcal{D} \setminus \mathcal{A}_0$.
- 6: **for** $r = 1, \dots, 5$ **do**
- 7: Infer required pool features; fit class statistics from $\mathcal{A} \setminus \mathcal{H}$ when needed.
- 8: Select b distinct candidates through capped passes, without accessing their labels.
- 9: Query the selected labels; update $\mathcal{A}$ and $\mathcal{U}$.
- 10: Begin the scheduled training phase; retain model and optimizer momentum.
- 11: **for** each class-balanced training step **do**
- 12: Optimize the task plus regularization loss; update training moments.
- 13: **if** 100 steps elapsed or this is the final phase step **then**
- 14: Measure loss on $\mathcal{H}$ and update the designated regularization controller.
- 15: **end if**
- 16: **end for**
- 17: Save and evaluate the final scheduled phase checkpoint.
- 18: **end for**
- 19: Evaluate the final checkpoint with the frozen full-set AutoAttack protocol.
- 20: **end for**

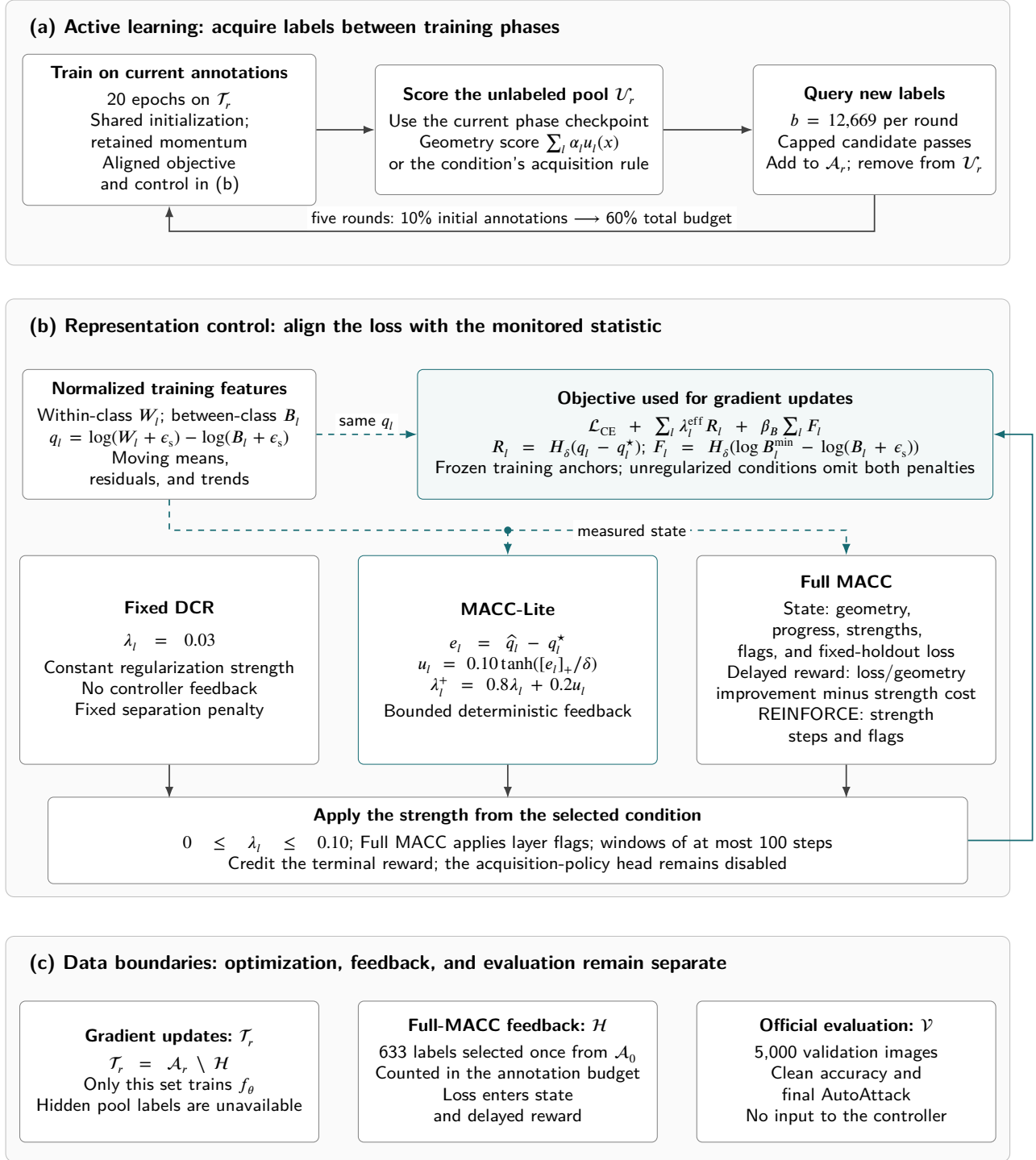
feature and retained-spectrum dimensions and, when necessary, the centered sample-rank bound $n_l - 1$. Matching the eigenvalues does not specify eigenvectors, class means, or an equiangular frame.

For diffusion, features are spatially pooled U-Net mid-block activations and the task loss is noise-prediction mean squared error. For audio, DCR acts on learned prefix embeddings and pooled audio-token-encoder features; the task loss is next-token cross-entropy. These experiments evaluate spectral regularization without learned label acquisition.

5. Supporting analysis

The analysis concerns the measured geometry, the effect of an isolated aligned gradient, coefficient bounds, and additional conditions connecting geometry to prediction. These are supporting characterizations, not global optimization, generalization, or input-space robustness guarantees for the training system.

Proposition 1 (Scatter decomposition and invariance). *For equal class weights and positive class counts, define $T_l =$*



Solid arrows: execution or applied control. Dashed arrows: measured signals.

Fixed DCR, MACC-Lite, and Full MACC are alternative experimental conditions.

Figure 1: The classification workflow of CÆNL. (a) Acquisition expands the annotated set between training phases. (b) The regularizer and adaptive controllers use the same normalized scatter statistic, while the separation safeguard penalizes insufficient between-class separation. Fixed strength, deterministic feedback, and learned feedback are compared as alternatives. (c) The fixed holdout supplies Full-MACC feedback without entering task-model gradient updates; official evaluation is separate. Language, diffusion, and audio use the task-specific spectral formulations. Prepared in TikZ with ChatGPT (OpenAI) assistance.

$K^{-1} \sum_c n_c^{-1} \sum_{i: y_i=c} \|h_{i,l} - \bar{\mu}_l\|_2^2$. Then $T_l = W_l + B_l$. Under Equation (1), $0 \leq T_l \leq 1$. The three moments and q_l are invariant under a common orthogonal transformation. Away from the norm floor, they are invariant under common positive rescaling of the raw features.

Proof. Expand $h_{i,l} - \bar{\mu}_l = (h_{i,l} - \mu_{c,l}) + (\mu_{c,l} - \bar{\mu}_l)$. The cross term vanishes within each class. Averaging gives $T_l = W_l + B_l$. Moreover,

$$T_l = \frac{1}{K} \sum_c \frac{1}{n_c} \sum_{i: y_i=c} \|h_{i,l}\|_2^2 - \|\bar{\mu}_l\|_2^2 \leq 1.$$

Orthogonal transformations preserve norms and inner products. Normalization cancels a common positive scale when neither representation activates the norm floor. $\square$

Lemma 2 (Quadratic deviation is not covariance trace). *Let Z have mean μ and covariance Σ , and let A be a fixed symmetric matrix. If second moments exist,*

$$\mathbb{E}[(Z - \mu)^\top A (Z - \mu)] = \text{tr}(A\Sigma). \quad (18)$$

Euclidean squared deviation therefore has expectation $\text{tr}(\Sigma)$, while squared Mahalanobis deviation with $A = \Sigma^{-1}$ has expectation d when Σ is nonsingular. With $(\Sigma + \epsilon I)^{-1}$, $\epsilon > 0$, the expectation is $\sum_j \sigma_j / (\sigma_j + \epsilon)$ rather than $\text{tr}(\Sigma)$.

Proof. Write the quadratic form as $\text{tr}(A(Z - \mu)(Z - \mu)^\top)$ and exchange expectation with the finite-dimensional trace. Diagonalizing Σ gives the special cases. $\square$

The identity does not establish that a high Mahalanobis score is an optimal acquisition choice. If A is estimated from the same random sample, the fixed-matrix argument requires conditioning or further analysis.

Proposition 3 (Direct local alignment). *Fix a measurement batch, a layer, and a positive residual $g(\theta) = q_l(\theta) - q_l^*$. Suppose g has an L_g -Lipschitz gradient, with $L_g > 0$, on a neighborhood containing the step. For $\eta, \lambda \geq 0$ and $a = H'_\delta(g(\theta)) \in [0, 1]$, the isolated step $\theta' = \theta - \eta\lambda a \nabla g(\theta)$ satisfies*

$$g(\theta') \leq g(\theta) - \eta\lambda a \left(1 - \frac{L_g \eta \lambda a}{2}\right) \|\nabla g(\theta)\|_2^2. \quad (19)$$

The bound is nonincreasing when $0 \leq \eta\lambda a \leq 2/L_g$.

Proof. Apply $g(\theta + v) \leq g(\theta) + \nabla g(\theta)^\top v + (L_g/2)\|v\|_2^2$ with $v = -\eta\lambda a \nabla g(\theta)$. $\square$

This result concerns one isolated loss component on a fixed batch. It does not imply monotone scatter reduction under the combined loss, momentum, augmentation, changing minibatches, or moving observations. It also does not establish that explicit objective–feedback alignment is necessary for useful control.

Proposition 4 (Bounded feedback coefficients). *If $0 \leq \lambda_{l,0} \leq \lambda_{\max}$ and $0 < \rho \leq 1$, Equation (13) keeps every coefficient in that interval. For a fixed residual, so that $u_{l,t} = u_l$,*

$$\lambda_{l,t} - u_l = (1 - \rho)^t (\lambda_{l,0} - u_l). \quad (20)$$

Proof. The desired coefficient lies in $[0, \lambda_{\max}]$. Each update is a convex combination of that value and the previous coefficient. For a fixed input, subtract u_l and iterate. $\square$

This establishes coefficient bounds and a fixed-input response. It does not establish stability of the coupled network–controller system. Full MACC’s projection supplies coefficient bounds but not this deterministic response formula.

Proposition 5 (Conditional nearest-centroid error bound). *Let population class means μ_c have minimum pairwise distance $d_{\min} > 0$, and assume equal class probabilities. A nearest-centroid classifier has error probability at most*

$$\min\{1, 4W/d_{\min}^2\}, \quad W = \frac{1}{C} \sum_c \mathbb{E}[\|H - \mu_c\|_2^2 \mid Y = c]. \quad (21)$$

Proof. A feature within distance $d_{\min}/2$ of its true mean is closer to that mean than to any other. An error therefore requires $\|H - \mu_Y\|_2^2 \geq d_{\min}^2/4$. Apply Markov’s inequality and average over classes. $\square$

This is a population result for a particular classifier. The trained softmax classifier need not use the nearest-centroid rule; a batch ratio is not a population bound; and aggregate $B_l > 0$ does not ensure separation of every pair. Reducing q_l alone therefore need not reduce this bound.

Input-space robustness. Let $m_{y,j}(x) = f_y(x) - f_j(x)$. If every margin is $L_{y,j}$ -Lipschitz on the relevant L_∞ ball, then $m_{y,j}(x) > \epsilon L_{y,j}$ for all $j \neq y$ is sufficient to preserve the correct class throughout that ball, because

$$m_{y,j}(x + v) \geq m_{y,j}(x) - L_{y,j} \|v\|_\infty.$$

Neither DCR formulation bounds those Lipschitz constants. Geometry and empirical robustness are therefore measured separately.

6. Experimental design

6.1. Study sequence and evidence separation

Engineering checks verify implementation behavior and are not counted as predictive-performance experiments. The original ImageNet-100 spectral pilot uses seeds 601 and 602. Saved-checkpoint attack diagnostics and a bounded aligned-objective development screen precede the final seeds 801–806. The aligned and spectral development protocols also differ in sampling, BatchNorm treatment, and integration choices; their absolute accuracies cannot isolate the effect of changing DCR.

The initial ImageNet-100 comparison evaluated nine conditions across six paired seeds. After reviewing those results, we evaluated two additional conditions: entropy acquisition with aligned DCR and MACC-Lite, and a matched-protocol UHerd-Margin comparator. Each used the original six

seed-specific initializations, targets, splits, annotation budgets, and evaluation endpoint. Their comparison lists were fixed before the additional runs, and their statistical families are reported separately. A third follow-up added entropy with fixed aligned DCR, retaining the same per-seed initial model and optimizer, splits, targets, floor penalty, acquisition protocol, and optimization schedule. Its two contrasts were declared before the new fixed-condition outcomes were inspected, after the entropy and Lite outcomes were already known. This later control adds six runs and 30 phases; it is not fresh-seed confirmation. Throughout this paper, a *condition* is a method configuration, a *run* is one condition under one seed, and a *phase* is one acquisition/training stage within a run. The resulting classification record comprises 72 method–seed runs and 360 post-initialization phases across 12 conditions; the number of independent paired seeds remains six.

C4 uses six separate training seeds and an additional unused-shard evaluation of the same saved checkpoints. Diffusion development uses seeds 1001 and 1002, followed by confirmation seeds 1101–1106. Audio development uses seeds 1201 and 1202 for two epochs per condition; confirmation uses seeds 1301–1306 for ten epochs. Confirmation starts from the original pretrained assets and new seed-specific task components, not development-final checkpoints. Only audio confirmation evaluates caption-model predictions on the retained official test split. Development results are not pooled into confirmation inference.

6.2. ImageNet-100 data and label accounting

We use the fixed 100-class subset associated with Contrastive Multiview Coding [17], drawn from ImageNet [18]. It contains 126,689 training images and 5,000 validation images, with 50 validation images per class. Original image bytes and dimensions are retained. A manifest records class mappings, source revision, paths, and SHA-256 values, which are checked before execution. The evaluated benchmark is ImageNet-100, not ImageNet-1K.

Stratified sampling supplies 12,669 initial annotations per seed. The controller holdout contains 633 of these, leaving 12,036 gradient-training images. Five rounds each acquire 12,669 additional examples. Annotation totals are 12,669, 25,338, 38,007, 50,676, 63,345, and 76,014. The final gradient-training set contains 75,381 images. The final annotation fraction is approximately 60.0005%, reported as 60%. Holdout labels are included in every method’s annotation cost. Pool labels are consulted only after selection.

6.3. Classification optimization and matching

The backbone is a randomly initialized ResNet-50 with a 100-class head [19]. Each seed has one shared 40-nominal-epoch cross-entropy initialization, reused by the original methods and later matched conditions. A physical minibatch contains 16 uniformly selected represented classes and eight distinct acquired examples per class. A nominal epoch has $\lceil |\mathcal{T}_r|/128 \rceil$ sampled minibatches rather than a permutation pass through the labeled set.

Training uses random resized 224×224 crops and horizontal flips. Scoring, feedback, and evaluation resize

the shorter side to 256 and take a 224-pixel center crop. ImageNet normalization is inside the model, so attacks act in pixel space. Augmentation and sampling streams are keyed by seed, phase, step, and example index. Common random-number conventions do not make sampled images identical once acquired sets differ.

Optimization uses SGD with momentum 0.9, Nesterov acceleration, and weight decay 10^{-4} on multidimensional parameters. One-dimensional parameters receive no decay. Parameters remain FP32 with BF16 autocast, and gradients are clipped at norm 5. Initialization uses learning rate 0.05, three nominal warm-up epochs, and cosine decay. Each subsequent 20-epoch phase starts at rate 0.025 with one warm-up epoch and cosine decay; momentum is retained. BatchNorm running statistics are updated during initialization and frozen afterward; affine parameters remain trainable.

The five acquired-label phases contain 3,880, 5,840, 7,820, 9,800, and 11,780 optimizer steps. Initialization contains 3,800 steps. Each conceptual condition run therefore includes 42,920 steps and 5,493,760 sampled example presentations. Initialization is executed once per seed and allocated once per conceptual condition when reporting comparable costs.

6.4. Baselines and acquisition cost

Table 2 separates acquisition, regularization, and adaptation. BADGE uses exact factorized last-linear-layer weight-gradient inner products rather than the pilot’s random projection; bias terms are excluded. Noise Stability uses five globally normalized Gaussian perturbations of the trainable parameter vector at relative scale 0.001, normalized softmax-output deviations, concatenation without projection, and greedy farthest-first selection. Model parameters and buffers are restored after noisy forwards.

Nonrandom methods cache required pool outputs at the current checkpoint. Mahalanobis acquisition additionally estimates class statistics from acquired examples. Equal annotation budgets and pass sizes do not equalize acquisition time, especially for Noise Stability. Random plus aligned fixed DCR tests whether the objective contributes without geometric acquisition. The final matrix has no new full-label reference; the pilot’s different-budget supervised reference is not mixed into the matched final comparison.

6.5. UHerding-Margin: matched-protocol implementation

The recent comparator implements the uncertainty-weighted coverage mechanism of Uncertainty Herding [13], as `uherding_margin_initial_r50`. Each seed’s shared-initial supervised ResNet-50 supplies unit-normalized layer4 features g_i , cached for the full training pool and fixed across rounds. This differs from a separately pretrained self-supervised representation. The task classifier continues under the common warm-start schedule.

Before acquisition, temperature T minimizes 15-bin ECE over 81 logarithmically spaced values in $[0.1, 10]$ using the budgeted 633-image holdout. Temperature-scaled

Table 2

Original frozen ImageNet-100 matrix. All nine conditions use seeds 801–806 and the same 10–60% annotation schedule. “None” disables both aligned penalties; all conditions record geometry. The three follow-up conditions are reported separately.

Condition	Acquisition	Representation control
Random	Uniform random selection	None
Entropy	Highest predictive entropy	None
BADGE	Exact factorized weight-gradient distances	None; independent implementation, bias excluded
Noise Stability	Perturbed-output deviations with greedy coverage	None; five noise draws, independent implementation
CÆNL acquisition only	Predicted-class diagonal Mahalanobis score	None
Random + aligned fixed DCR	Uniform random selection	Aligned objective, $\lambda_l = 0.03$
CÆNL + aligned fixed DCR	Mahalanobis score	Aligned objective, $\lambda_l = 0.03$
CÆNL + aligned MACC-Lite	Mahalanobis score	Bounded, smoothed coefficient feedback
CÆNL + aligned Full MACC	Mahalanobis score	Learned coefficient increments and activation flags

probabilities define $\omega_i = 1 - (p_{i,(1)} - p_{i,(2)})$. The bandwidth σ is the smallest positive Euclidean distance between fixed features of currently annotated examples, including held-out annotations. Squared distances at most 10^{-12} are numerical zero. The kernel is $k_\sigma(g_i, g_j) = \exp(-\|g_i - g_j\|_2^2 / \sigma^2)$.

For candidate pass P , greedy selection maximizes

$$G_P(j \mid C) = \sum_{i \in P} \omega_i \left[k_\sigma(g_i, g_j) - \max_{a \in C} k_\sigma(g_i, g_a) \right]_+, \quad (22)$$

where C contains annotated centers and examples already selected in the round. The 10,000-candidate and 10%-per-pass convention is retained until 12,669 new examples are acquired. Distances, kernels, and gains use float64. Lazy greedy is checked against exhaustive greedy on deterministic CPU and CUDA fixtures.

No DCR is applied. Initializations, optimizer states, label accounting, balanced batches, post-initialization BatchNorm treatment, and the final attack endpoint match the task protocol. Feature extraction, acquisition including calibration, and training are timed separately. This comparison evaluates the disclosed adaptation, not the original authors’ representation, unrestricted selection domain, or published benchmark scores.

6.6. Classification evaluation and statistical analysis

Every completed phase is evaluated on all 5,000 official validation images in full precision. Reported outcomes include pooled cross-entropy, top-1 and top-5 accuracy, 15-bin ECE, and equal-class normalized moments. The selected checkpoint is the final scheduled checkpoint, not the best validation epoch. Logits, predictions, labels, and compact class-moment sufficient statistics are retained.

The final checkpoint receives standard AutoAttack [14] at L_∞ radius 8/255 on all 5,000 images. Its components are APGD-CE, targeted APGD, targeted FAB, and Square Attack, without shortened prescribed iterations. Attack minibatches contain 64 images; restartable storage chunks contain 256. Pixel bounds, perturbation norms, and survival masks are

recorded. No adversarial training is used. A compatibility repair changed view to reshape in the library’s zero-gradient diagnostic, retaining the diagnostic, attack configuration, and completed training.

The paired experimental units are six seeds, conditional on the fixed dataset and protocol. Seven contrasts are specified separately for final clean and robust accuracy. For target-minus-reference differences d_i , the individual paired- t interval is

$$\bar{d} \pm t_{0.975,5} \frac{s_d}{\sqrt{6}}.$$

Both paired- t tests and enumeration of all 2^6 sign flips are retained. Holm correction [20] is applied within each declared endpoint and test family. Intervals are individual, unadjusted intervals, not simultaneous confidence bands. The paired- t analysis relies on its distributional approximation for the differences. Interpreting sign-flip enumeration as an exact null test requires the corresponding sign-exchangeability or symmetry assumption; enumeration alone does not make it assumption-free.

Resolution of the six-seed exact tests. For the two-sided absolute-mean statistic, an observed sign pattern and its global negation are both at least as extreme as the observed absolute mean. Therefore

$$p_{\text{exact}} \geq \frac{2}{2^6} = 0.03125. \quad (23)$$

In a Holm family of $m \geq 2$ comparisons at $\alpha = 0.05$, the first rejection threshold is $0.05/m \leq 0.025$. Consequently, no rejection is attainable under these six-seed, two-sided, Holm-adjusted exact tests. A smallest raw value of 0.03125 expresses the strongest attainable sign-flip evidence at this resolution, not adjusted significance. All declared tests and adjustments are reported with this resolution limit; it is not evidence that their underlying effects are absent.

The entropy-plus-Lite, UHerding, and entropy-plus-fixed follow-ups each use a separate two-contrast clean-accuracy

Table 3

Task-specific regularization and feedback conventions. “Lite” and “Full” denote different task-specific implementations, not an unchanged controller transferred across all tasks. Task protocols provide the remaining settings.

Task	Loss and observed statistic	Fixed/adaptive coefficients	Update convention
ImageNet-100 classification	Aligned normalized within/between log ratio; fixed separation safeguard	Fixed 0.03; adaptive initialization 0.03; bounds [0, 0.10]	Aligned Lite uses the leaky rule; Full uses the aligned reward; windows of at most 100 steps and phase-boundary handling
GPT-2/C4	Spectral loss with $k = 128$, ranks 16 and 4 at h6/h12; separate covariance-spectrum feedback	Fixed 0.01; adaptive initialization 0.005; bounds [0, 0.08]	Spectral control every 25 steps; Lite step size 0.002 and deadband 0.1
CIFAR-10 diffusion	Spectral loss with $k = 64$, rank-16 target; full EMA-covariance monitor	Fixed 0.01; adaptive initialization 0.01; bounds [0, 0.05]	Spectral control every 50 steps; Lite step size 0.001 and deadband 0.1
AudioCaps captioning	Spectral loss with $k = 64$, rank-16 targets at prefix/encoder; different sample axes and zero padding	Fixed 0.01; adaptive initialization 0.01; bounds [0, 0.05]	Spectral control every 50 steps and final step; terminal reward credited without an unused next action

family, specified before the respective additional runs but after earlier outcomes were available. For UHerd, the contrasts are UHerd minus entropy and entropy-plus-Lite minus UHerd. For the fixed-control follow-up, they are entropy-plus-fixed minus entropy and entropy-plus-Lite minus entropy-plus-fixed. New fixed-condition robustness is descriptive; the original robustness family is unchanged. These within-family corrections do not provide one family-wise error guarantee for the full sequential research programme. Geometry, calibration, top-5 accuracy, and intermediate budgets are descriptive secondary outcomes.

6.7. Cross-task experimental protocols

These protocols concern Equation (16). The fixed, Lite, and Full names identify task-specific implementations; the aligned classification loss is not used in these studies.

6.7.1. Language modeling on C4

Pretrained GPT-2 [21], with 124,439,808 parameters, is fine-tuned on a deterministic C4 subset [22]. Each of seeds 301–306 processes 99,999,744 tokens in 12,207 steps, with sequence length 1,024 and eight sequences per step. Baseline, fixed spectral DCR, spectral MACC-Lite, and spectral Full MACC share the budget and training configuration. The spectral implementation monitors h6 and h12 with target ranks 16 and 4, retained spectrum size 128, and a 2,048-token DCR subsample. Fixed strength is 0.01.

Training uses the first English C4 training shard, with the first validation shard used for feedback and development (Appendix B). All four conditions therefore evaluate the same single-shard training distribution. The original reporting set has 1,889 validation sequences, with 64 additional sequences reserved for controller feedback.

After checkpoint selection and training were frozen, all 24 checkpoints were evaluated on validation shard 00002, which

had not been used for local training, feedback, or configuration development. Each checkpoint makes 1,999,872 next-token predictions. NLL and accuracy are pooled over tokens; ECE is computed from pooled confidence-bin statistics. Perplexity is the exponential of NLL, not an independent endpoint. “Untouched” concerns local study use, not verified absence of GPT-2 pretraining overlap.

Optimization uses learning rate 5×10^{-5} , weight decay 0.01, and 200 warm-up steps. AdamW has $\beta_1 = 0.9$, $\beta_2 = 0.95$; one-dimensional, bias, and normalization parameters are exempt from decay. Norm-1 clipping and linear warm-up followed by cosine decay toward 10% of the base rate are used. Parameters remain FP32 with BF16 CUDA autocast.

Spectral controllers operate every 25 steps. Both adaptive methods initialize their coefficients at 0.005 within [0, 0.08]. Spectral Lite uses step size 0.002 and deadband 0.1. Full MACC uses two 128-unit hidden layers, policy learning rate 3×10^{-4} , exploration probability 0.15 decreasing to 0.02, reward-baseline weight 0.1, and entropy coefficient 0.001. Its coefficient increments are $\{-0.005, -0.0025, 0, 0.0025, 0.005\}$. Its language-specific reward settings include validation-delta scale 5.0, value-delta clip 0.1, target-deviation weight 0.005, and zero query penalty. These settings are not replaced by the later diffusion/audio or aligned-classification controller settings.

The unused-shard evaluator compares fixed DCR, Lite, and Full separately with baseline, giving three contrasts per endpoint. It reports individual paired- t intervals, raw paired- t and exact sign-flip values, and Holm adjustment of the paired- t values within each three-contrast family. Holm adjustment is applied separately to the paired- t and exact values within each endpoint family. NLL is primary; perplexity is its monotone transformation, while token accuracy and ECE remain secondary.

6.7.2. CIFAR-10 diffusion

We fine-tune the 35,746,307-parameter pretrained goog1e/ddpm-cifar10-32 U-Net with the noise-prediction objective [23]. Seeds 1101–1106 evaluate baseline fine-tuning, fixed spectral DCR, spectral MACC-Lite, and spectral Full MACC. The unchanged pretrained model is also sampled with each seed’s generation-noise stream. Those six references measure sampling variation for one model, not six pretrained-model replications.

A fixed partition of the original 50,000 CIFAR-10 training images [24] supplies 45,000 fine-tuning images and 5,000 internal holdout images. The holdout contains 256 controller images and 1,024 disjoint denoising-assessment images with fixed noise and timesteps. These images may have been seen during pretraining; the holdout is not pristine relative to the pretrained model. Official test outcomes do not enter control or checkpoint selection.

Each trained condition performs 10,000 updates with batch size 128, matched per-seed image order and noise streams, horizontal flips, AdamW learning rate 2×10^{-5} , zero weight decay, and norm-1 clipping. Parameters are FP32 with BF16 autocast. EMA weights use decay 0.9995. The final scheduled checkpoint is retained independently of performance. No active label acquisition occurs.

DCR uses 256-dimensional spatially pooled mid-block features across timesteps 0–999, retained spectrum size 64, and target rank 16. The full EMA-covariance monitor uses decay 0.99 and cadence 50 steps. Fixed strength is 0.01. Adaptive methods start at 0.01 within [0, 0.05]. For finite monitored rank $r_{\text{eff},l,t}$, spectral Lite uses

$$e_{l,t}^{\text{sp}} = \left| \log \left(\frac{\max\{r_{\text{eff},l,t}, 10^{-12}\}}{16} \right) \right| - 0.1, \quad \lambda_{l,t+1} = \Pi_{[0,0.05]} \left(\lambda_{l,t} + 0.001 e_{l,t}^{\text{sp}} \right), \quad (24)$$

where $\Pi_{[a,b]}(v) = \min\{b, \max\{a, v\}\}$. The error is not replaced by its positive part: the coefficient increases outside the log-rank deadband, decreases inside it, and is unchanged at its boundary, subject to projection. Missing or nonfinite rank observations leave the corresponding coefficient unchanged. This accumulating projected rule differs from the leaky classification rule in Equation (13).

Spectral Full MACC uses the retained two-layer 128-unit policy, increments $\{-0.005, -0.0025, 0, 0.0025, 0.005\}$, activation flags, policy learning rate 3×10^{-4} , exploration decreasing from 0.10 to 0.02, baseline weight 0.05, and entropy coefficient 0.001. Its reward combines the scaled held-out loss improvement (scale 10, clipped to $[-2, 2]$) and a $-0.1 |\log(r_{\text{eff}}/16)|$ target-deviation term. The query head is disabled, and uniformly explored actions are excluded from on-policy gradient updates. At termination, the pending action is credited without sampling an unused next action.

Each final EMA model generates 50,000 uint8 images with DDIM [25], 100 sampling steps, $\eta = 0$, and batches of 128. Noise streams are matched within a seed. The primary endpoint is FID [26] against the complete original 50,000-image training distribution, using torch-fidelity 0.4.0.

KID [27] and Inception score [28] are secondary. A final EMA denoising assessment uses all 10,000 official test images and fixed noise/timestep seed 541211. Per-image errors are retained.

Four FID contrasts are frozen: baseline minus pretrained reference and each regularized condition minus baseline. Negative differences favor the target. Paired- t and all 64 sign flips are reported with separate Holm correction across these four contrasts. The exact-resolution limitation in Section 6.6 applies. Development uses two different seeds, 2,000 updates, and 5,000 generated samples; its absolute FID values are not pooled with confirmation.

6.7.3. AudioCaps captioning

The audio study uses original AudioCaps annotations [29] and a documented available-recording subset from OpenSound/AudioCaps. All 473 hosted Parquet shards were checked against published digests. IDs, start times, and captions were matched to pinned original annotations. The retained counts are 44,466 training, 440 validation, and 866 test clips, versus original inventories of 49,838, 495, and 975 unique clips. Coverage is 89.22%, 88.89%, and 88.82%. Each retained validation/test clip has five references.

Preparation excludes recordings outside eight to twelve seconds, one test identity with conflicting decoded variants, and both members of a nonzero waveform duplicate spanning train and test. Sources, variants, and exclusion records are preserved. Four all-zero training waveforms remain flagged and retained. Checks found no cross-split video IDs or exact nonzero waveform duplicates, including after actual mono/resampling preprocessing. These checks do not establish acoustic authenticity, absence of near-duplicates, or equivalence to an official raw-audio distribution. The mirror’s provenance and licensing limitations remain explicit; raw audio is not redistributed.

The frozen Audio Spectrogram Transformer (AST) [30] checkpoint MIT/ast-finetuned-audio-10-10-0.4593 supplies shared features. Channels are averaged, resampled to 16 kHz, and limited to the first 10.24 seconds for a single-window representation. Original waveforms are not rewritten. FP32 AST computation supplies 64 pooled patch tokens of dimension 768, stored in FP16 and reused read-only. The test features use the frozen backbone only. AudioSet [31] pretraining may include underlying recordings, so no absence of pretraining overlap is claimed.

The task model has a two-layer trainable audio-token encoder, an eight-token learned prefix, and trainable pretrained GPT-2, totaling 141,774,336 parameters. Baseline, fixed spectral DCR, spectral Lite, and spectral Full use matched accepted initial task weights, epoch permutations, per-step dropout seeds, and optimizer budgets. Seeds 1301–1306 start from original GPT-2 assets and new seed-specific encoder/prefix parameters, not pilot-final weights. The initialization recovery is documented in Appendix B.

AdamW uses encoder/prefix rate 10^{-4} , decoder rate 5×10^{-5} , decay 0.01, and norm-1 clipping. Parameters remain FP32 with BF16 autocast. Each epoch permutes 44,466

training pairs and uses 1,389 batches of 32, dropping the last 18 pairs. A new deterministic permutation is used every epoch. Each of the 24 runs performs ten epochs, or 13,890 updates.

For zero-based optimizer-step index $s = 0, \dots, S-1$, with $S = 13,890$ and $w = 100$ warm-up steps, every parameter group's base rate $\eta_{g,0}$ is multiplied by

$$a(s) = \begin{cases} (s+1)/w, & 0 \leq s < w, \\ \frac{1 + \cos(\pi p(s))}{2}, & w \leq s, \end{cases} \quad (25)$$

where $p(s) = \min\{1, (s-w)/(S-w)\}$ and $\eta_g(s) = \eta_{g,0}a(s)$. The schedule has terminal factor zero and no positive learning-rate floor. Its zero endpoint is $s = S$; the last executed update is $s = S-1$, so its final applied factor is slightly above zero. The final scheduled checkpoint is evaluated without best-epoch selection or early stopping.

Spectral sample axes and rank. The prefix representation reshapes eight embeddings per clip into $32 \times 8 = 256$ vectors of dimension 768. The pooled encoder representation has 32 vectors of dimension 768. Thus their centered sample-rank bounds are 255 and 31, respectively. The configured 64-entry spectrum is a storage and truncation dimension, not a claim of 64 nonzero eigenvalues for both representations. The encoder Gram spectrum is zero-padded to 64 entries as in Equation (17). A rank-16 template is feasible for both full training batches. The EMA-covariance monitor is distinct from either current-batch spectrum and can have a larger effective rank.

Control operates every 50 steps and at the final step, yielding 278 adaptive events. The fixed coefficient is 0.01. Adaptive coefficients start at 0.01 within $[0, 0.05]$. Spectral Lite uses Equation (24) at both monitored layers, with step size 0.001, deadband 0.1, and target rank 16. Full retains the diffusion spectral-policy settings, replacing denoising feedback loss with caption loss on the audio holdout. Its terminal action is credited without an unused next action, and the acquisition head is disabled.

The first 64 validation IDs in frozen manifest order provide controller feedback; the remaining 376 provide development assessment. Neither is used for task-model gradient updates. Final confirmation checkpoints evaluate all 866 retained test recordings. Beam size is three, decoding batch size is 32, and at most 30 new tokens are generated.

The primary endpoint is CIDEr-D [32] from the retained pycocoevalcap/PTB path. BLEU-1–4 [33], ROUGE-L [34], METEOR [35], SPICE [36], SPIDEr [37], and pooled token NLL are secondary outcomes. SPIDEr combines CIDEr-D and SPICE on their recorded scales. NLL is weighted by actual next-token target counts. A separate development diagnostic mismatches audio and captions deterministically without training updates.

The three primary CIDEr-D contrasts compare fixed, Lite, and Full with baseline. Positive differences favor the regularized condition. Both paired- t and all 64 sign flips are

retained with separate Holm corrections across these three comparisons and individual unadjusted intervals. Secondary outcomes do not replace the primary endpoint. The six paired seeds remain the experimental units; repeated evaluation of 866 clips does not create extra training replications.

6.8. Execution environment and reproducibility

Experiments run sequentially on an IBM Cloud instance with one NVIDIA L40S GPU, PyTorch 2.6.0, torchvision 0.21.0, CUDA runtime 12.4, and Python 3.11.7. Datasets and checkpoints reside on persistent storage. Protocols, source hashes, splits, acquisitions, controller records, summaries, and attack-chunk digests are retained. Integrity checks establish consistency of recorded artifacts, not third-party replication of GPU execution.

6.9. Computational assistance and figure preparation

ChatGPT (OpenAI) assisted with experimental-software development and review, saved-result analysis, mathematical exposition, reference checking, and LaTeX/TikZ/Matplotlib preparation. Numerical figures are compiled from recorded measurements or explicitly stated statistical summaries; schematic figures describe the implemented workflow. Numerical outputs are derived from the recorded experiments and reproducible calculations. The verification scope is described in Appendix B.

7. Results and analysis

7.1. ImageNet-100 comparison across matched seeds

Table 4 consolidates all 12 evaluated conditions. Entropy acquisition with aligned DCR and MACC-Lite has the highest mean clean accuracy, $73.360 \pm 0.208\%$, compared with $71.463 \pm 0.320\%$ for entropy alone and $71.747 \pm 0.626\%$ for UHerding-Margin. The later entropy-plus-fixed condition reaches $72.263 \pm 0.381\%$. The table distinguishes the

initial comparison from the three follow-up conditions; their inferential families remain separate. Within the initial nine-condition comparison, geometric acquisition with aligned MACC-Lite attains $72.120 \pm 0.490\%$. Adding fixed aligned DCR to geometric acquisition gives $+0.850$ percentage points, and replacing fixed strength with Lite gives another $+1.080$ points. Both component differences are positive in all six paired seeds.

Table 4

All 12 ImageNet-100 conditions at 60% annotations. Values are mean $\pm$ sample SD over six paired seeds; clean and standard AutoAttack accuracy use all 5,000 validation images. Study I denotes the initial nine-condition comparison, A the later acquisition-rule ablation, B the later UHerding comparison, and C the later entropy-plus-fixed control. The table is descriptive: study-specific inferential families remain separate. Bold identifies the highest observed mean clean accuracy, not a universal superiority claim.

Condition	Study	Clean top-1 (%)	AutoAttack 8/255 (%)	Cross-entropy	ECE
Random	I	69.293 ± 0.334	0.013 ± 0.010	1.272712 ± 0.021145	0.094945 ± 0.002917
Entropy	I	71.463 ± 0.320	0.010 ± 0.011	1.150591 ± 0.020293	0.079698 ± 0.003849
BADGE	I	71.110 ± 0.467	0.003 ± 0.008	1.149581 ± 0.007022	0.077088 ± 0.004058
Noise Stability	I	71.023 ± 0.249	0.003 ± 0.008	1.162821 ± 0.006167	0.078703 ± 0.002375
CÆNL acquisition only	I	70.190 ± 0.240	0.013 ± 0.010	1.210545 ± 0.010404	0.085552 ± 0.004501
Random + aligned fixed DCR	I	70.113 ± 0.452	0.017 ± 0.008	1.217939 ± 0.021315	0.085964 ± 0.006228
CÆNL + aligned fixed DCR	I	71.040 ± 0.427	0.017 ± 0.008	1.155351 ± 0.015230	0.078428 ± 0.003667
CÆNL + aligned MACC-Lite	I	72.120 ± 0.490	0.007 ± 0.010	1.078885 ± 0.008014	0.062948 ± 0.004965
CÆNL + aligned Full MACC	I	70.803 ± 0.318	0.007 ± 0.010	1.171322 ± 0.023746	0.081639 ± 0.004671
Entropy + aligned MACC-Lite	A	73.360 ± 0.208	0.003 ± 0.008	1.028967 ± 0.010101	0.055426 ± 0.001256
UHerding-Margin	B	71.747 ± 0.626	0.007 ± 0.010	1.127922 ± 0.021039	0.073977 ± 0.003294
Entropy + aligned fixed DCR	C	72.263 ± 0.381	0.000 ± 0.000	1.092014 ± 0.013386	0.071037 ± 0.004253

Figure 2 shows the available per-phase clean-accuracy measurements, separating acquisition rules from aligned regularization. The budget trajectories are descriptive; the

final 60% endpoint remains the basis of the declared clean-accuracy comparisons. The entropy-plus-Lite and entropy-plus-fixed trajectories are shown separately in Figure 4, together with paired gains over entropy. All displayed points are measured phase endpoints.

The two component contrasts pass the planned adjusted paired- t threshold. Lite versus entropy does not: its raw paired- t value is 0.0466, but its adjusted value is 0.0931. The individual interval narrowly excludes zero but is not a simultaneous interval. Figure 3 preserves the original effects and intervals. The nonrejection of the adjusted exact tests must be interpreted with their design-imposed resolution limit, not as an independent demonstration that the effects are absent.

Acquisition alone improves over random by $+0.897$ points, while random plus fixed aligned DCR improves by $+0.820$. These comparisons show favorable directions for both components within the declared regime. Acquisition alone remains below entropy in mean accuracy. The acquisition-rule ablation shows that geometric acquisition is not necessary for the strongest observed result. No equal-performance full-supervision comparison establishes a 40% annotation saving.

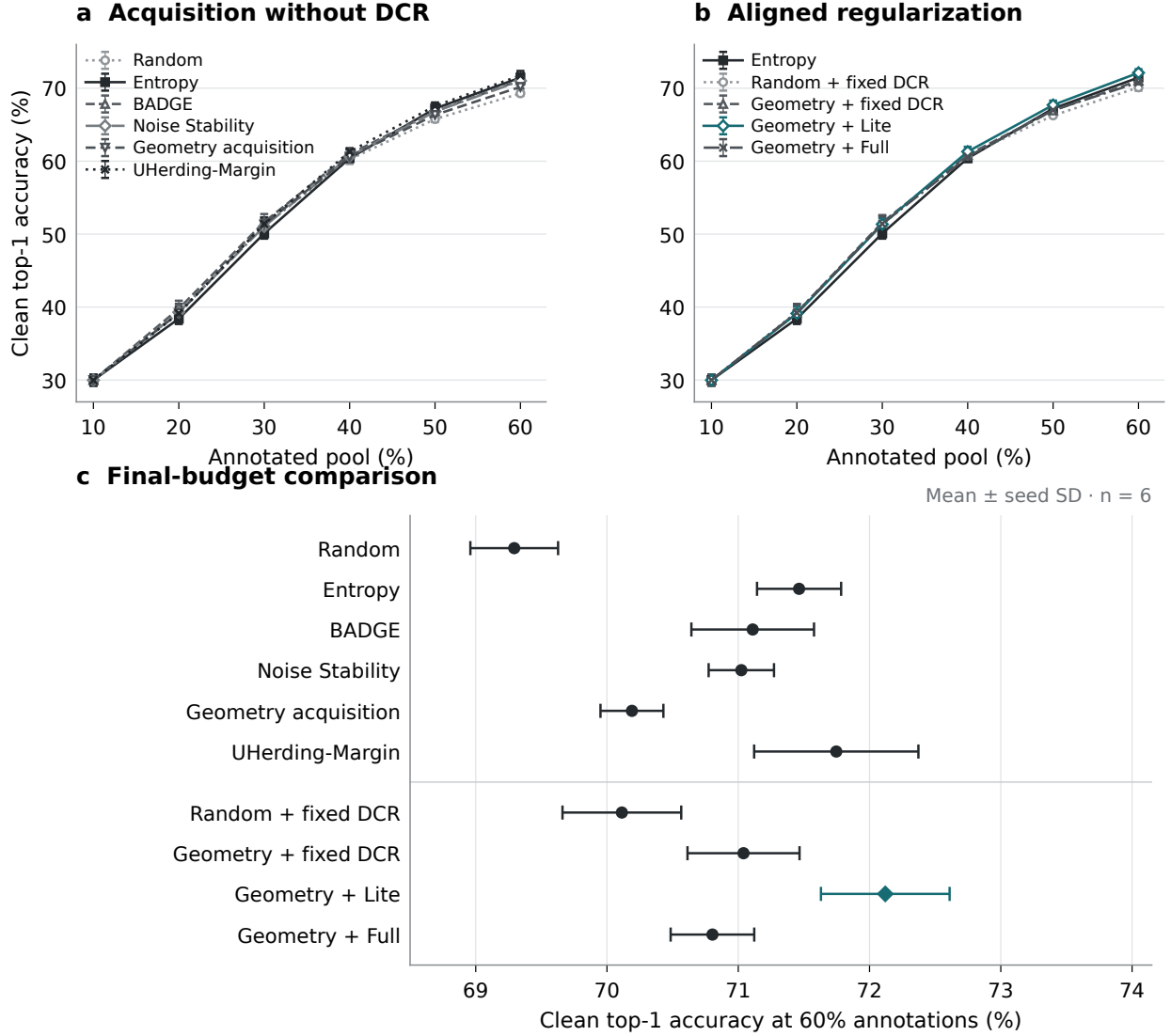


Figure 2: Acquisition and aligned regularization under the common ImageNet-100 protocol. Panels (a) and (b) show measured accuracy trajectories on identical axes; panel (c) enlarges the final 60% endpoint for the original nine conditions and the UHerding-Margin follow-up. Markers and bars represent six-seed means and sample SDs, not confidence intervals. All trajectories start from the same 10% checkpoint. Fixed, Lite, and Full denote aligned DCR conditions; entropy regularization follow-ups appear in Figure 4. Connecting lines guide the eye, and no new area-under-curve test or inferential family is introduced.

Table 5

All seven original clean-accuracy contrasts in percentage points. Intervals are individual unadjusted 95% paired- t intervals. Each p -value column is separately Holm-adjusted across seven contrasts. The exact tests have the resolution limit described in Section 6.6.

Target minus reference	Difference (pp)	95% CI (pp)	Adjusted t p	Adjusted exact p
Aligned fixed DCR – acquisition only	+0.850	[+0.430, +1.270]	0.0208	0.2188
Aligned MACC-Lite – aligned fixed DCR	+1.080	[+0.447, +1.713]	0.0309	0.2188
Aligned Full MACC – aligned fixed DCR	−0.237	[−0.714, +0.241]	0.2589	0.2813
Aligned MACC-Lite – entropy	+0.657	[+0.015, +1.299]	0.0931	0.2188
Aligned Full MACC – entropy	−0.660	[−0.869, −0.451]	0.0032	0.2188
Acquisition only – random	+0.897	[+0.389, +1.405]	0.0309	0.2188
Random + aligned fixed DCR – random	+0.820	[+0.297, +1.343]	0.0309	0.2188

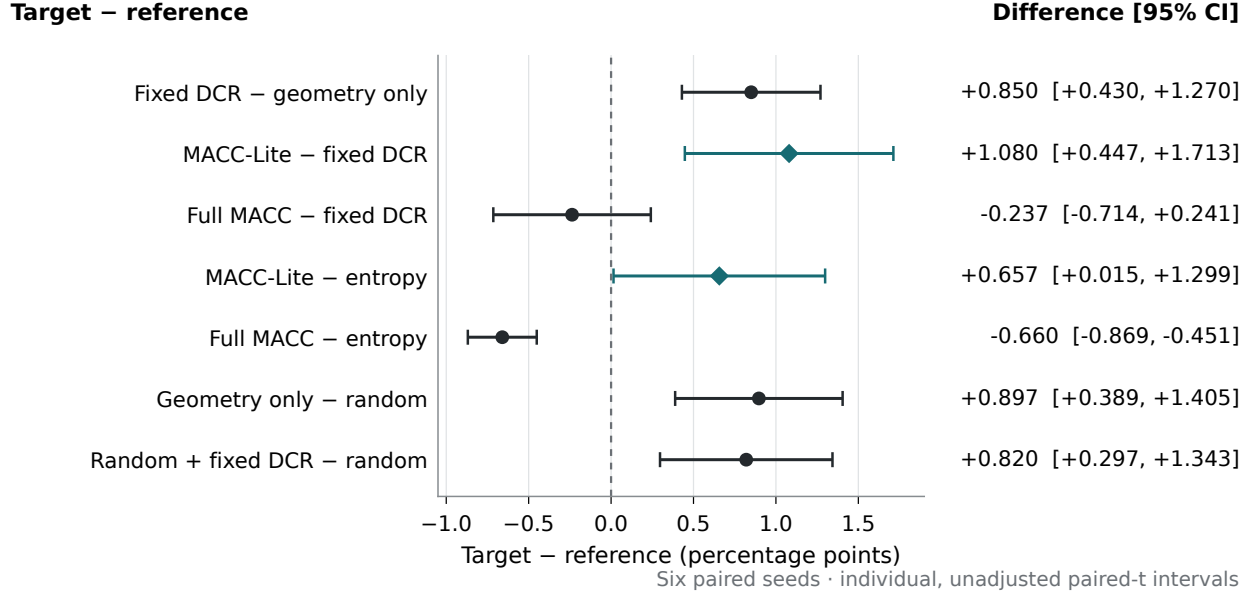


Figure 3: All seven planned clean-accuracy contrasts in the original nine-condition family. Points give the mean target-minus-reference difference; bars and the numeric column give individual unadjusted 95% paired- t intervals across six seeds. The dashed line denotes no difference. Fixed DCR, MACC-Lite, and Full MACC use geometry acquisition unless random acquisition is explicitly indicated. Accent color identifies Lite contrasts, not statistical significance. Multiplicity-adjusted tests are reported in Table 5; Lite versus entropy does not pass the adjusted paired- t threshold.

7.2. Acquisition-rule ablation under aligned MACC-Lite

The acquisition-rule condition replaces geometric acquisition with entropy while retaining the aligned objective, Lite settings, six shared initializations, annotation and feedback splits, targets, and five 20-epoch phases. Its first acquisition is checked against the original entropy selection. Thirty additional phases and full-set final attacks are completed. These are paired acquisition-rule runs, not new independent seeds for the original matrix.

Entropy with aligned DCR and Lite reaches $73.360 \pm 0.208\%$ and exceeds unregularized entropy and geometry-plus-Lite in every seed. Table 6 gives the paired comparisons; Appendix A provides the seed-level outcomes (Table 19).

The gain over entropy is +1.897 percentage points, with individual 95% paired- t interval [+1.625, +2.169]. The gain over geometry plus Lite is +1.240 points, with interval [+0.707, +1.773]. Both comparisons pass the acquisition-rule adjusted paired- t analysis. Their raw exact value is 0.03125 and adjusted value 0.0625, consistent with the resolution limit. The evidence supports the *combined aligned-loss-and-Lite treatment within entropy acquisition*. That acquisition-rule study did not include a fixed-strength entropy control. The later matched follow-up in Section 7.3 provides that comparison while preserving the acquisition-rule study’s original two-contrast family.

One image survives the attack for seed 802 and none for the other seeds, giving mean robust accuracy 0.0033%. This descriptive result does not change the robustness conclusion or the original robustness family.

Table 6

Two acquisition-rule clean-accuracy contrasts. Positive values favor entropy with aligned DCR and Lite. Intervals are individual unadjusted paired- t intervals; each test family is Holm-adjusted across these two contrasts. This family was specified after inspection of the initial comparison.

Reference condition	Gain (pp)	95% CI (pp)	Adjusted t p	Adjusted exact p
Entropy without aligned regularization	+1.897	[+1.625, +2.169]	0.0000198	0.0625
Geometry acquisition + aligned MACC-Lite	+1.240	[+0.707, +1.773]	0.001875	0.0625

7.3. Matched fixed-strength control within entropy acquisition

The third follow-up keeps entropy acquisition and replaces Lite with the fixed aligned coefficient $\lambda_l = 0.03$ at layer3 and layer4. The separation penalty remains $\beta_B = 0.01$

with the original per-seed targets and floors. All six initial model and optimizer states, label/holdout splits, training settings, and the five-round schedule are reused. The first acquisition matches original entropy exactly; later acquired sets are allowed to change with the trained model. This tests

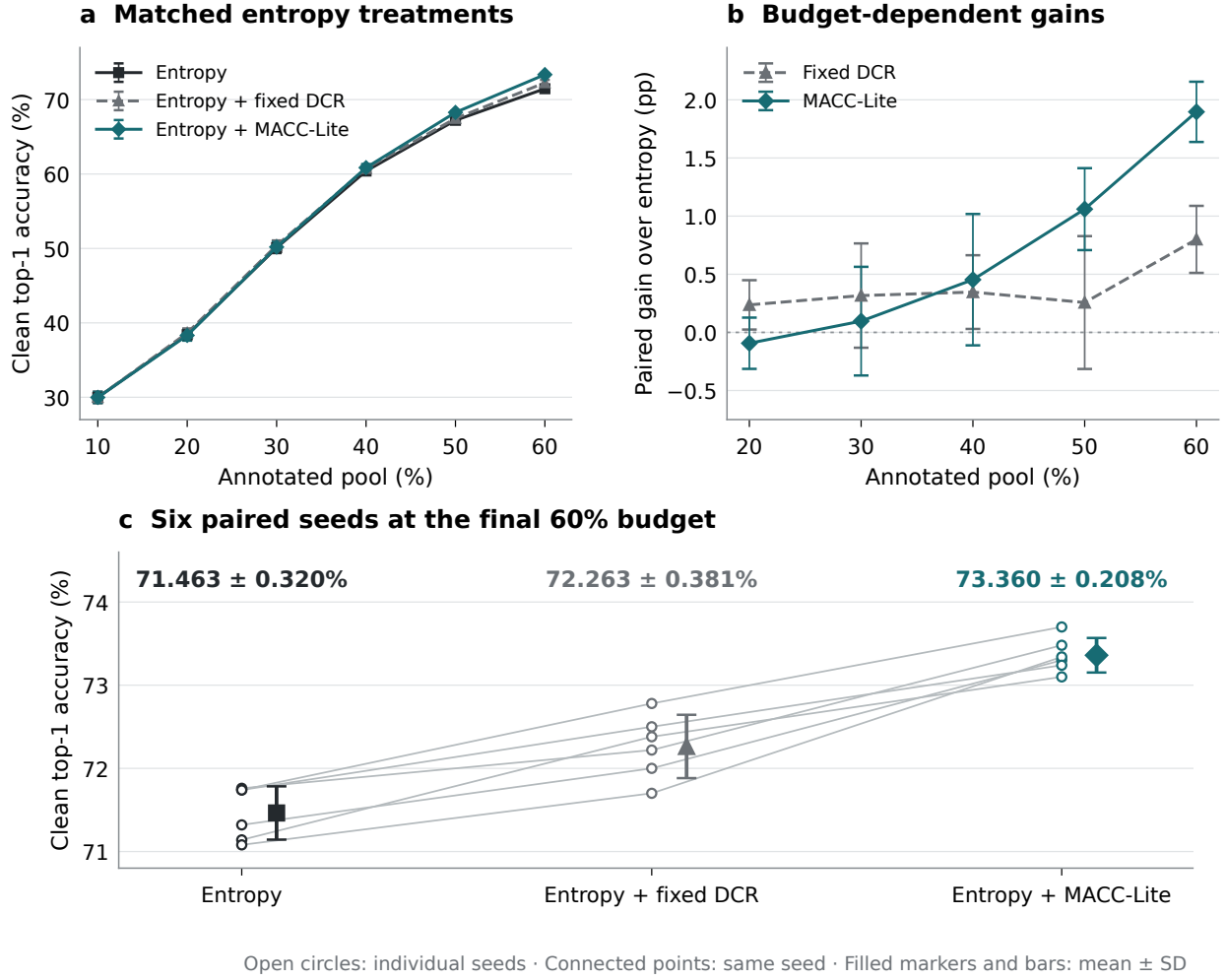


Figure 4: Matched entropy treatments across annotation budgets and seeds. (a) Mean clean accuracy and sample SD across six seeds. (b) Mean and sample SD of within-seed differences from entropy; the common 10% initialization has zero gain and is omitted. (c) Individual final-budget accuracies, with lines joining the same seed across conditions; adjacent filled markers and bars show mean $\pm$ sample SD. Overlapping seed values are retained. These bars are not confidence intervals. Fixed DCR uses $\lambda_l = 0.03$ and the same separation penalty as Lite. Lite is not uniformly better at early budgets, but exceeds both comparators in all six final-budget seeds. Only the final 60% comparisons enter the declared follow-up tests. All points derive from recorded measurements.

the coefficient-control treatment within the active-learning pipeline, including its subsequent effect on selection.

Entropy plus fixed DCR reaches $72.263 \pm 0.381\%$, improving over entropy by $+0.800$ percentage points with individual 95% interval $[+0.498, +1.102]$. Lite improves over fixed by $+1.097$ points with interval $[+0.714, +1.479]$. Both differences are positive in all six seeds. Within this new two-contrast family, both Holm-adjusted paired- t values are 0.001442. Both raw exact values are 0.03125 and adjusted values are 0.0625. Table 7 reports the full statistics, and Table 19 gives the paired outcomes. The later study supplies

evidence favoring Lite over the specified fixed coefficient; it does not establish superiority over a tuned coefficient or a prespecified open-loop schedule.

Figure 4 includes the Lite trajectory and the new fixed trajectory. At 20% and 30% annotations, fixed DCR has slightly higher mean accuracy than Lite; the ordering reverses at later budgets. Thus the final-budget advantage should not be described as uniform throughout acquisition. Fixed DCR has zero robust-correct images in every seed under standard AutoAttack 8/255 on all 5,000 validation images. The new condition does not alter the near-zero robustness conclusion.

Table 7

The later fixed-control study's two clean-accuracy contrasts. Differences and individual unadjusted 95% paired- t intervals are in percentage points. Holm correction is applied separately to the two paired- t and the two exact tests in this family. Prior families are unchanged.

Target minus reference	Difference	95% CI	Raw t p	Adjusted t p	Exact raw / adjusted
Entropy + fixed – entropy	+0.800	[+0.498, +1.102]	0.001047	0.001442	0.03125 / 0.0625
Entropy + Lite – entropy + fixed	+1.097	[+0.714, +1.479]	0.000721	0.001442	0.03125 / 0.0625

7.4. Comparison with UHerding-Margin

UHerding-Margin completes all five phases for each seed and the full-set final evaluation. Its mean clean accuracy is $71.747 \pm 0.626\%$, versus $71.463 \pm 0.320\%$ for entropy and $73.360 \pm 0.208\%$ for entropy with aligned DCR and Lite. UHerding minus entropy is +0.283 points with interval $[-0.430, +0.997]$, adjusted paired- t value 0.3540, and adjusted exact value 0.3125. This does not establish superiority or equivalence.

Entropy with the aligned Lite treatment exceeds the UHerding adaptation in every seed. The mean difference is

+1.613 points, with interval $[+0.957, +2.270]$. The adjusted paired- t value is 0.002923; the adjusted exact value is 0.0625. This compares complete pipelines and does not establish superiority of entropy alone, or of the combined treatment over the authors' unmodified UHerding benchmark implementation.

UHerding retains one correct adversarial image for each of seeds 802 and 805, and none otherwise: mean robust accuracy 0.0067%. The comparison adds an executed recent mechanism while retaining its representation and selection-domain adaptations.

Table 8

The UHerding comparison's two clean-accuracy contrasts, in percentage points. Intervals are individual unadjusted 95% paired- t intervals; each test column is Holm-adjusted across these two comparisons only.

Target minus reference	Difference	95% CI	Adjusted t p	Adjusted exact p
UHerding-Margin – entropy	+0.283	$[-0.430, +0.997]$	0.3540	0.3125
Entropy + aligned Lite – UHerding-Margin	+1.613	$[+0.957, +2.270]$	0.002923	0.0625

7.5. Representation changes and learned-controller behavior

Within the original matrix, geometry with aligned Lite has the lowest mean log trace ratio, cross-entropy, and ECE.

Table 9 reports descriptive geometry separately from task performance. These statistics are not a causal mediation analysis, and the all-class assessment ratio does not establish attainment of the 16-class anchor target.

Table 9

Final all-class normalized geometry, mean $\pm$ seed SD. The statistic averages layer3 and layer4 $\log((W + \epsilon)/(B + \epsilon))$.

Condition	Mean log trace ratio
Random	1.3881 ± 0.0197
Entropy	1.3916 ± 0.0170
Acquisition only	1.3864 ± 0.0151
Random + aligned fixed	1.0607 ± 0.0218
CÆNL + aligned fixed	0.9767 ± 0.0200
CÆNL + aligned Lite	0.4937 ± 0.0390
CÆNL + aligned Full	1.0825 ± 0.0982
Entropy + aligned fixed	1.0027 ± 0.0125
Entropy + aligned Lite	0.4725 ± 0.0434

Full MACC executes 370, 368, 368, 374, 365, and 373 learned-policy updates for seeds 801–806. Its mean accuracy is 0.237 points below aligned fixed DCR, with an interval spanning zero, and 0.660 points below entropy. The latter difference passes the adjusted paired- t test. The learned controller is active, but its additional capacity does not establish benefit under the present settings. Feedback horizon, changing acquisition phases, and imperfect credit assignment are possible explanations rather than identified causes.

Short development continuations found Lite coefficients below the 0.10 cap. That observation does not substitute for a

long-run stability analysis. Proposition 4 bounds coefficients, not the network's geometric state or task error.

7.6. Adversarial robustness is not recovered

Final checkpoints retain zero or one correct adversarial prediction out of 5,000. Mean robust accuracy is approximately 0.000%–0.017%. No planned robustness comparison establishes an improvement. These near-zero empirical outcomes are not a population certificate, and repeated predictions on the same images do not become independent observations when pooled across seeds.

The original spectral pilot also had zero robustness at 8/255; extending its selected-checkpoint diagnostic from 1,000 to all 5,000 images reproduced that result. Smaller-radius post-hoc diagnostics did not establish an overall advantage over random acquisition. The final study retains the original attack radius. Improved clean performance and supervised geometry therefore do not support the earlier robustness-benefit interpretation.

7.7. C4: a small spectral-regularization effect

Table 10 reports the unused local validation shard, and Table 11 gives all three NLL contrasts against baseline. Fixed spectral DCR lowers NLL in every paired seed, with mean difference -0.00029250 nats/token and individual interval $[-0.00032813, -0.00025688]$. Its raw paired- t value is 4.42×10^{-6} and its archived within-endpoint Holm-adjusted

value is 8.85×10^{-6} . The raw exact value is 0.03125; applying Holm correction within the same three-contrast family gives 0.09375.

The effect is close to the original-shard difference of -0.00028734 . The second shard checks sensitivity to local data reuse but adds no independent training seeds. The practical change is approximately 0.0086% in NLL and 0.0292% in perplexity. Fixed DCR does not establish a token-accuracy or pooled-ECE benefit. Spectral Lite increases NLL in all six seeds by a mean of 0.00194729 nats/token, with interval $[0.00187898, 0.00201560]$. Full is inconclusive relative to baseline: $+0.00023624$, interval $[-0.00018778, +0.00066025]$, raw paired- t value 0.2115. This is not an equivalence result. The secondary comparison outputs are retained in Appendix C.

Table 10

Unused-local-shard C4 evaluation of the same 24 final GPT-2 checkpoints. Values are mean $\pm$ sample SD over six training seeds, with 1,999,872 next-token predictions per checkpoint. The formulations are spectral, not aligned classification DCR.

Condition	NLL (nats/token)	Perplexity	Pooled token ECE	Token accuracy
Baseline	3.384212 ± 0.000239	29.49473 ± 0.00705	0.012324 ± 0.000426	0.380067 ± 0.000123
Fixed spectral DCR	3.383919 ± 0.000253	29.48611 ± 0.00746	0.012216 ± 0.000438	0.380029 ± 0.000083
Spectral MACC-Lite	3.386159 ± 0.000220	29.55222 ± 0.00651	0.013131 ± 0.000396	0.379920 ± 0.000098
Spectral Full MACC	3.384448 ± 0.000462	29.50170 ± 0.01362	0.012438 ± 0.000584	0.380040 ± 0.000111

Table 11

All three unused-shard C4 NLL contrasts. Differences and individual 95% paired- t intervals are in 10^{-4} nats/token; negative differences favor regularization. Holm adjustment is within the three NLL contrasts, separately for each test. Both test families use the same three contrasts; intervals are unadjusted.

Condition minus baseline	Difference	95% CI	Raw t p	Holm t p	Raw exact p	Holm exact p
Fixed spectral DCR	-2.92505	$[-3.28129, -2.56881]$	4.4239×10^{-6}	8.8477×10^{-6}	0.03125	0.09375
Spectral MACC-Lite	$+19.47289$	$[+18.78979, +20.15599]$	8.9651×10^{-9}	2.6895×10^{-8}	0.03125	0.09375
Spectral Full MACC	$+2.36236$	$[-1.87778, +6.60250]$	0.21152	0.21152	0.21875	0.21875

The language controller records 488 actions per seed. Lite spends much of training at the 0.08 coefficient cap; Full records 428–445 learned-policy updates. The adaptive results therefore do not reflect an inactive controller. Effective ranks increase under regularization, so the mechanism is spectral shaping rather than uniformly stronger low-rank collapse. The small fixed-DCR NLL result does not establish broader calibration, memorization, or cross-modal gains.

Table 12

Diffusion confirmation with 50,000 generated images per condition and seed. Values are mean $\pm$ sample SD over six seed streams. FID is primary. Reference variation is sampling variation for one unchanged pretrained model.

Condition	FID $\downarrow$	KID $\times 10^3 \downarrow$	Inception score $\uparrow$
Pretrained reference	10.512 ± 0.100	9.153 ± 0.124	8.479 ± 0.039
Unregularized fine-tuning	8.057 ± 0.048	6.605 ± 0.084	8.692 ± 0.043
Fixed spectral DCR	8.213 ± 0.070	6.726 ± 0.105	8.724 ± 0.049
Spectral MACC-Lite	8.472 ± 0.100	6.939 ± 0.115	8.728 ± 0.055
Spectral Full MACC	8.211 ± 0.116	6.744 ± 0.088	8.729 ± 0.050

7.8. Diffusion: fine-tuning helps, spectral regularization does not

Unregularized fine-tuning lowers mean FID from 10.512 to 8.057. Fixed spectral DCR, Lite, and Full instead increase FID relative to that baseline by 0.156, 0.415, and 0.154. Each increase occurs in every seed. Improvement over the unchanged pretrained model therefore does not establish added value from DCR.

Table 13

All four diffusion FID contrasts. Positive differences indicate worse FID for the target. Intervals are individual unadjusted paired- t intervals. Each test family is Holm-adjusted over these four comparisons.

Target minus reference	Difference	95% interval	Adjusted t p	Adjusted exact p
Baseline – pretrained reference	–2.45468	[–2.58346, –2.32589]	2.6769×10^{-7}	0.125
Fixed DCR – baseline	+0.15601	[+0.11357, +0.19845]	0.0004481	0.125
MACC-Lite – baseline	+0.41472	[+0.31931, +0.51012]	0.0003004	0.125
Full MACC – baseline	+0.15431	[+0.06205, +0.24658]	0.0077204	0.125

All four contrasts pass their adjusted paired- t tests. Each raw exact value is 0.03125, giving adjusted value 0.125. The consistent adverse direction is retained with the small-sample assumptions and resolution qualification. It is not a claim that every diffusion regularizer is harmful.

KID is also higher in mean under every regularized condition. Slightly higher Inception scores remain secondary and do not replace FID. Mean official-test EMA denoising MSE is 0.02918539 for baseline, 0.02918606 for fixed DCR, 0.02918634 for Lite, and 0.02918632 for Full, compared with 0.02935395 for the reference. These small descriptive differences are not formal equivalence or significant secondary effects.

Full executes 188, 186, 189, 185, 183, and 186 learned updates, with 200 controller events per seed. Lite also executes 200 events and ends at the 0.05 cap. The final value does not specify its occupancy over training. Final monitored ranks are approximately 23–24 for baseline, 49 for fixed

DCR, 38–39 for Lite, and 42–60 for Full. Active control and representation changes do not establish target attainment or better generation.

The two-seed pilot favored fixed DCR and Full relative to its baseline. That direction does not persist in confirmation. Training horizon and generated-sample count both changed, so the present comparison does not isolate the reason for the reversal.

7.9. Audio captioning: no established CIDEr-D improvement

Fixed DCR has a slightly higher mean CIDEr-D than baseline; Lite and Full have lower means. Every primary interval spans zero, and both adjusted test families give $p = 1.0000$ for all three contrasts. This is an inconclusive comparison, not proof of equivalence, zero population effect, or universal ineffectiveness.

Table 14

AudioCaps results on the fixed available subset: mean $\pm$ sample SD over seeds 1301–1306. Each run uses 866 test clips with five references each. Metrics retain their native scales; CIDEr-D is primary.

Condition	CIDEr-D	BLEU-4	METEOR	SPICE	SPIDER
Baseline	0.6736 $\pm$ 0.0220	0.2401 $\pm$ 0.0137	0.2328 $\pm$ 0.0043	0.1664 $\pm$ 0.0046	0.4200 $\pm$ 0.0132
Fixed spectral DCR	0.6745 $\pm$ 0.0185	0.2384 $\pm$ 0.0114	0.2335 $\pm$ 0.0022	0.1650 $\pm$ 0.0041	0.4198 $\pm$ 0.0110
Spectral MACC-Lite	0.6668 $\pm$ 0.0212	0.2378 $\pm$ 0.0108	0.2320 $\pm$ 0.0035	0.1664 $\pm$ 0.0020	0.4166 $\pm$ 0.0112
Spectral Full MACC	0.6703 $\pm$ 0.0175	0.2382 $\pm$ 0.0094	0.2319 $\pm$ 0.0023	0.1659 $\pm$ 0.0041	0.4181 $\pm$ 0.0100

Table 15

All three primary audio CIDEr-D contrasts. Positive differences favor the target. Intervals are individual unadjusted paired- t intervals; each test family is Holm-adjusted across three comparisons.

Target minus baseline	Difference	95% interval	Adjusted t p	Adjusted exact p
Fixed spectral DCR	+0.000896	[–0.016040, +0.017833]	1.0000	1.0000
Spectral MACC-Lite	–0.006781	[–0.028950, +0.015389]	1.0000	1.0000
Spectral Full MACC	–0.003329	[–0.015313, +0.008656]	1.0000	1.0000

Seed-level CIDEr-D values are reported in Appendix A.

Raw paired- t values are 0.8971, 0.4673, and 0.5072; raw exact values are 0.84375, 0.50000, and 0.53125. Unlike a case resting only on the exact-test resolution limit, these raw results also provide no strong evidence of a difference. Mean test NLL is 2.026833 for baseline, 2.029867 for fixed, 2.033260 for Lite, and 2.030362 for Full. These remain descriptive secondary results.

Full records 260, 253, 260, 266, 259, and 254 learned-policy updates. Both adaptive conditions record 278 events.

All 24 runs produce 866 nonempty captions. The mismatched-audio development diagnostic has higher NLL than correctly paired audio in every run. That establishes sensitivity to the supplied conditioning under this diagnostic, not added benefit from DCR or a comprehensive grounding guarantee.

The pilot evaluates 376 development clips after two epochs; confirmation evaluates 866 test clips after ten epochs. Their absolute scores cannot attribute a difference solely to training duration. The study evaluates audio–text captioning; audio retrieval, audio classification, other architectures, and

full-original-benchmark comparability remain outside its scope.

7.10. Computational cost and memory

Table 16 reports ImageNet-100 components. The synchronized training-compute timer includes forward, loss, backward, clipping, and optimizer execution, but excludes

subsequent controller observation and policy operations. Feedback and acquisition have separate timers. Loading, serialization, evaluation, and orchestration are not fully represented by summing these columns. All references record geometry, and initialization is allocated once per conceptual condition.

Table 16

Recorded ImageNet-100 timing components in seconds per conceptual condition run, averaged over six seeds. Training and feedback include their allocated shared-initialization costs. These are component, not complete application, timings.

Method	Training compute	Feedback inference	Acquisition	Final AutoAttack
Random	5093.24	182.25	0.85	792.52
Entropy	5092.81	182.45	490.83	800.95
BADGE	5093.97	182.55	523.26	775.80
Noise Stability	5093.61	182.52	2942.89	777.36
Acquisition only	5093.23	182.38	739.69	798.50
Random + aligned fixed	5240.51	182.30	0.86	811.51
CÆNL + aligned fixed	5238.40	182.28	740.18	820.73
CÆNL + aligned Lite	5238.85	182.27	739.58	804.70
CÆNL + aligned Full	5240.13	182.30	738.97	783.91

The maximum PyTorch-allocated memory is 6,075,090,432 bytes for every conceptual run in the original nine-condition matrix. Shared initialization may dominate that maximum; equal values do not establish zero additional

controller memory. The phase-specific comparison below is narrower but still does not isolate a complete controller footprint.

Table 17

Post-initialization training time and peak allocation. Duration is the mean sum over five post-initialization training phases. Peaks describe PyTorch allocation in those phases, not isolated controller memory.

Condition	Five-phase duration (s)	Recorded phase peak (bytes)
Entropy	5,255.02	6,066,439,680
Entropy + aligned MACC-Lite	5,391.31	6,066,701,824
Difference	+136.29	+262,144

The duration in Table 17 and the synchronized training-compute component in Table 16 have different measurement boundaries. The former is a phase-duration record; the latter excludes controller observation and policy operations and includes an allocated initialization component. They are not additive or directly subtractable. These timers do not define a common end-to-end boundary.

The entropy-plus-Lite mean phase duration is 2.59% higher, a difference of 136.29 s. These are sequential campaigns, not interleaved timing trials. The peak difference is 262,144 bytes, or 0.25 MiB. Neither quantity is complete application overhead or an isolated controller measurement.

The entropy-plus-fixed follow-up separately records means of 4,837.85 s for synchronized training compute, 182.55 s for feedback inference, 483.99 s for acquisition, and 782.66 s for final AutoAttack. The training and feedback components exclude the reused initialization; they therefore cannot be inserted directly into Table 16 or compared as complete application time. The fixed-control records do not contain a phase-duration or phase-peak memory aggregate for this new condition, so no new overhead ratio is inferred.

A five-pair GPT-2/C4 microbenchmark times 250 optimizer steps. Plain execution takes 33.5293 ± 0.0429 s; the

sampled-monitoring path takes 33.7633 ± 0.0731 s. The paired time increase is 0.698% with SD 0.142 percentage points; peak allocation increases about 0.089 GiB. Its BF16 parameters, microbatches of two sequences with four accumulation passes, ten monitor events, optimizer initialization, and timing boundary differ from production.

Production C4 time increases by 7.88%, 7.68%, and 16.62% for fixed, Lite, and Full relative to its monitored baseline. These ratios cannot be combined with the microbenchmark to obtain total overhead relative to plain training.

Diffusion mean task-job wall times are 1,616.81 s for reference sampling and 2,919.80, 2,974.48, 2,978.48, and 2,977.43 s for baseline, fixed, Lite, and Full. They include task-driver training, sampling, and metrics, but exclude the wrapper's later official-test EMA assessment and study preparation. The reference has no training updates and is not an overhead denominator.

Audio successful training-loop times span 1,175.53–1,201.80 s for baseline, 1,308.80–1,333.21 s for fixed, 1,309.24–1,337.74 s for Lite, and 1,336.39–1,375.73 s for Full. These include controller feedback and checkpoints but exclude setup, final decoding, metrics, and saved-result

audits. Feature extraction is reused from the pilot. Rejected

8. Discussion and limitations

8.1. What objective alignment contributes

The principal classification finding is a favorable *combined aligned-loss-and-Lite treatment* within the evaluated training regime. The original matrix supplies matched fixed-versus-none and Lite-versus-fixed comparisons under geometric acquisition. The acquisition-rule ablation supplies a favorable combined-treatment comparison and shows that the geometry score is unnecessary for the strongest observed mean.

The later entropy-plus-fixed control separates two observed treatment contrasts: +0.800 points for adding the fixed aligned loss and safeguard, and a further +1.097 points for replacing the fixed coefficient with Lite. Their sum recovers the +1.897-point combined-treatment difference. This arithmetic is a decomposition of matched pipeline contrasts, not a causal mediation analysis of geometry, feedback, or selected examples. The adaptive policy can also change subsequent acquired sets.

The fixed comparator uses $\lambda_l = 0.03$. A broader claim for feedback should address an independently tuned fixed

initialization attempts and verification are separate. No universal sub-10% or end-to-end overhead claim follows.

strength or a prespecified open-loop schedule with comparable tuning resources. The experiments do not separately isolate normalization and the separation safeguard. Attribution specifically to objective-feedback alignment would require a controlled comparison that changes alignment while matching the remaining optimization and sampling choices. The favorable final-budget contrast should also be distinguished from the mixed early-budget ordering in Figure 4.

Full MACC receives feedback and updates its policy in every seed, but its task outcomes are inconclusive or unfavorable. Its outcome is conditional on the specified state, reward, action, and horizon. A different controller might perform differently, but the present results do not establish why additional policy capacity fails to improve the task.

The UHerding comparison is similarly conditional on its fixed shared-initial supervised features, capped candidate passes, holdout temperature calibration, and warm-start training. The combined entropy-plus-Lite result is not a general rejection of uncertainty-weighted coverage.

Table 18 summarizes the empirical scope.

Table 18

Supported observations and claims not established. These distinctions apply to the reported protocols rather than all possible implementations of the framework.

Question	Supported observation	Not established
Classification alignment	Loss and feedback share the supervised log trace ratio	Necessity, global stability, or global optimality
Active learning	Matched six-seed budget comparison and favorable component directions	Universal superiority or equal-performance 40% label savings
MACC-Lite	Favorable Lite-versus-fixed contrasts under geometric and entropy acquisition	Superiority to tuned fixed strengths or schedules, isolated alignment causality, or necessary geometric acquisition
Full MACC	Learned regularization control executes in all seeds	A task benefit from extra controller capacity or learned query thresholds
Adversarial robustness	Full-set standard AutoAttack at the specified radius	Meaningful robustness gains, certification, or deployment safety
Language modeling	Small fixed spectral-DCR loss reduction on an unused local shard	An aligned-classification-objective effect or universal language gains
Diffusion	All spectral conditions have higher FID than matched fine-tuning	Universal generative benefit or a test of aligned classification DCR
Audio captioning	Six-seed available-subset test comparison without established CIDEr-D gain	Equivalence, full-original-benchmark comparability, or general audio gains

8.2. Limits of the classification and statistical evidence

The visual study uses one backbone, one 100-class subset, a class-balanced sampler, and frozen post-initialization BatchNorm statistics. Matching methods within this regime does not make it their best possible training recipe. Official validation images were used for reporting during development; new seeds quantify initialization and split variation, not transfer to an entirely untouched image benchmark.

Geometric atypicality may prioritize unhelpful examples. Aggregate scatter can conceal a poorly separated class pair,

and coefficient bounds do not establish stable optimization. Training anchors and all-class assessment also differ in class support and transforms. There is no demonstrated advantage at the specified adversarial endpoint or evidence of deployment safety.

The exact-test nonrejections have the structural limitation derived in Section 6.6. The adjusted paired-*t* findings remain conditional on their small-sample assumptions. Neither procedure removes the post-hoc chronology of the additional

comparisons or provides programme-wide multiplicity control. We retain the original comparison families instead of reorganizing them after observing the outcomes.

8.3. Cross-task scope and generalization limits

C4 concerns a small effect on a locally selected single-shard subset with pretrained GPT-2. Its unused-shard evaluation does not establish absence of pretraining overlap. Its spectral result cannot be attributed to the different aligned classification objective.

Diffusion evaluates one pretrained CIFAR-10 U-Net, one sampler, and one training schedule. Every spectral condition has worse FID than the matched fine-tuning baseline. This does not test aligned classification DCR on diffusion, a separately tuned regularization family, or large-scale text-to-image generation.

Audio captioning is conditional on an approximately 89%-coverage third-party subset. Exact identity and waveform checks do not prove acoustic authenticity or exclude every near-duplicate. AST pretraining may overlap underlying recordings, and four flagged zero training clips remain in the frozen policy. The study provides neither full-original-benchmark comparability nor broader audio-retrieval, classification, or grounding guarantees. Its inconclusive primary contrasts do not establish statistical equivalence.

The completed protocols thus cover classification, language, image generation, and audio captioning, but support no universal cross-task gain. ImageNet-1K and ViT performance remain unevaluated. The initialization workaround also limits claims of seed-only byte-identical reproduction, as documented in Appendix B.

8.4. Reproducibility and data availability

The retained record includes protocols, source snapshots, environment versions, splits, annotation accounting, controller histories, checkpoint digests, and evaluation outputs. Historical outcomes are preserved rather than overwritten by later variants. Appendix B summarizes data provenance and verification boundaries.

Code, experiment configurations, numeric result records, and statistical reconstruction scripts are available at <https://github.com/alqithami/CANL>. The repository distinguishes

the historical pipeline from the later aligned classification and spectral-confirmation implementations. It includes the manuscript source and a reproducible analysis entry point. Raw licensed data and large model checkpoints are excluded. The initialization and verification limits in Appendix B remain applicable; source availability is not a claim of independently repeated training.

9. Conclusion

Objective-aligned representation control improves classification in the specified ImageNet-100 protocol. Entropy acquisition with aligned DCR and MACC-Lite reaches $73.360 \pm 0.208\%$ accuracy at 60% annotations and exceeds matched fixed strength by 1.097 percentage points. This supports the tested adaptive treatment; it does not establish superiority over tuned fixed schedules or meaningful adversarial robustness. Full MACC adds no established classification benefit. The separate spectral studies yield a small C4 loss reduction, worse diffusion FID, and inconclusive audio effects. Representation control should therefore be judged by task performance, with its objective, feedback rule, computational cost, and evaluation scope made explicit.

Funding

This research received no external funding or in-kind support.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Saad Alqithami: Conceptualization, Methodology, Data curation, Formal analysis, Investigation, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review and editing.

A. Seed-level results and development studies

Tables 19, 20, and 21 report the paired observations underlying the main comparisons. The experimental units and statistical families are defined in Section 6.6.

Table 19

Matched entropy treatments at 60% annotations. Clean accuracies are percentages; differences are percentage points. Entropy and Lite are reused completed results. Lite robust-correct counts use standard AutoAttack 8/255 on all 5,000 validation images. Fixed DCR has 0/5,000 robust-correct images in every seed.

Seed	Entropy	Entropy + fixed	Entropy + Lite	Fixed – entropy	Lite – fixed	Lite robust correct
801	71.14	72.38	73.10	+1.24	+0.72	0/5,000
802	71.76	72.22	73.48	+0.46	+1.26	1/5,000
803	71.32	72.00	73.30	+0.68	+1.30	0/5,000
804	71.74	72.78	73.70	+1.04	+0.92	0/5,000
805	71.08	71.70	73.34	+0.62	+1.64	0/5,000
806	71.74	72.50	73.24	+0.76	+0.74	0/5,000
Mean	71.463	72.263	73.360	+0.800	+1.097	—
Sample SD	0.320	0.381	0.208	0.288	0.364	—

Table 20

Matched UHerding comparison at 60% annotations. Entropy references are reused completed results. UHerding uses the fixed-initial-R50 adaptation of Section 6.5. Robust counts refer to the new UHerding branches only.

Seed	Entropy	UHerding-Margin	Entropy + aligned Lite	UHerding robust correct
801	71.14	71.58	73.10	0/5,000
802	71.76	72.64	73.48	1/5,000
803	71.32	72.38	73.30	0/5,000
804	71.74	71.46	73.70	0/5,000
805	71.08	71.40	73.34	1/5,000
806	71.74	71.02	73.24	0/5,000
Mean $\pm$ SD	71.463 $\pm$ 0.320	71.747 $\pm$ 0.626	73.360 $\pm$ 0.208	—

Table 21

All six paired audio CIDEr-D values from the final confirmation test evaluation.

Seed	Baseline	Fixed	Lite	Full
1301	0.699902	0.706662	0.695297	0.697474
1302	0.683473	0.656485	0.683658	0.675072
1303	0.634166	0.656615	0.658886	0.647493
1304	0.681798	0.678712	0.663435	0.671009
1305	0.672890	0.676179	0.633839	0.655220
1306	0.669446	0.672400	0.665875	0.675435

A.1. What the development studies established

The aligned formulation was selected after a bounded mechanism screen. Each source seed had a shared 200-step stabilization and five matched 2,000-step continuations on a fixed labeled set. The screen required a favorable geometric direction and an assessment-loss increase no larger than 0.02 nats in either seed. This was a development tolerance, not statistical noninferiority.

Relative to no regularization, spectral DCR changed the geometric-mean trace ratio by +0.077% and +0.054%. Aligned fixed DCR changed it by −2.009% and −2.365%; aligned Lite by −4.426% and −5.015%; and aligned Full by −1.691% and −1.300%. Loss changes were mixed but within the screen’s tolerance. These findings motivated the subsequent end-to-end comparison; they did not themselves establish annotation efficiency or robustness.

B. Data provenance and verification scope

The repository accompanying this paper records the dataset revisions, source-file digests, experiment configurations, and exported per-seed measurements. Classification uses the CMC ImageNet-100 class selection from the original ImageNet images, preserved without reencoding. The selected inventory contains 126,689 training and 5,000 validation images. Locally recorded checksums establish consistency of these copies; they are not independent upstream certification.

C4 training uses `en/c4-train.00000-of-01024.json.gz`; the controller and development split uses `en/c4-validation.00000-of-00008.json.gz`. The final holdout evaluation uses `en/c4-validation.00002-of-00008.json.gz`. The second shard was unused in local training and development; absence of GPT-2 pretraining overlap is not established.

The original classification audit reconstructs metrics from saved validation, feedback, and attack arrays. Subsequent report-level checks reproduce the paired effects and statistical tests across classification, C4, diffusion, and audio, including the entropy-treatment budget curves and layer-geometry summaries. This verifies the arithmetic of exported records. It does not independently regenerate predictions, adversarial examples, generated images, or captions from raw inputs and checkpoints. The server-recorded audio audits additionally reproduce caption metrics from saved predictions and original references.

Audio initialization required a bounded retry for a pre-training mismatch in the encoder positional parameter. A construction was accepted only if its whole-model digest matched the original seed-specific record; no training outcome or evaluation score entered that acceptance rule. Completed outputs, seeds, and acceptance records were preserved. This procedure enforces the recorded match but does not resolve the underlying initialization nondeterminism. Accepted-state hashes alone cannot reconstruct their tensors, and the availability of all corresponding initial-state binaries is not established. Exact regeneration from seeds alone is consequently not claimed.

Raw licensed images, corpus text, audio recordings, and large model checkpoints are not distributed in the repository. Original data access conditions and the audio subset's coverage and provenance limitations continue to apply. The released code and numeric records permit statistical reconstruction within these stated boundaries.

C. Secondary C4 comparison statistics

The unused-shard analysis reports three method–baseline contrasts for each secondary endpoint. Table 22 retains the effect estimates, individual intervals, and both raw tests, with Holm correction within each endpoint. Its exact-adjustment column is calculated from the archived raw exact values for consistent presentation. Perplexity is descriptive rather than an independent primary outcome. Positive differences are adverse for perplexity and ECE; negative differences are adverse for token accuracy.

Table 22

Complete secondary unused-shard C4 comparisons. Each panel has its own three-contrast Holm family. Confidence intervals are individual and unadjusted. The indicated scaling applies to both differences and intervals; token ECE and token accuracy are proportions, not percentages.

Condition minus baseline	Difference	95% CI	Raw t p	Holm t p	Raw exact p	Holm exact p
<i>Perplexity: differences in 10^{-2}</i>						
Fixed spectral DCR	−0.86260	[−0.96757, −0.75763]	4.4061×10^{-6}	8.8122×10^{-6}	0.03125	0.09375
Spectral MACC-Lite	+5.74906	[+5.54778, +5.95034]	8.8781×10^{-9}	2.6634×10^{-8}	0.03125	0.09375
Spectral Full MACC	+0.69705	[−0.55390, +1.94800]	0.21147	0.21147	0.21875	0.21875
<i>Pooled token ECE: differences in 10^{-4}</i>						
Fixed spectral DCR	−1.07333	[−4.14827, +2.00161]	0.41069	0.78043	0.37500	0.68750
Spectral MACC-Lite	+8.07857	[+6.41836, +9.73877]	5.7942×10^{-5}	0.00017	0.03125	0.09375
Spectral Full MACC	+1.14058	[−1.97737, +4.25853]	0.39021	0.78043	0.34375	0.68750
<i>Token accuracy: differences in 10^{-4}</i>						
Fixed spectral DCR	−0.38336	[−1.21617, +0.44945]	0.28989	0.28989	0.28125	0.28125
Spectral MACC-Lite	−1.47509	[−1.89933, −1.05086]	0.00029	0.00088	0.03125	0.09375
Spectral Full MACC	−0.26752	[−0.63924, +0.10420]	0.12355	0.2471	0.09375	0.18750

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

ChatGPT (OpenAI) was used to assist with manuscript organization and editing, mathematical exposition, programming and debugging, interpretation of saved experimental artifacts, reference checking, and preparation of the LaTeX figures. Research-process uses and verification limits are described in Section 6.9 and Appendix B. The author is responsible for the scientific design, interpretation, and content of this work.

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